# Agronomic Nitrogen Efficiency of Bio-based Fertilizers for Crops

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# Abstract

The efficient use of bio-based fertilizers (BBFs) is achieved by a correct estimation on mineral N release potential of certain BBFs during the growing season.

This study aims to explore the potential influencing factors on BBFs bioavailability and plant N use efficiency by conducting a step-by-step systematic literature review. The potential effects of BBFs application on mineral N release within 90 days was investigated by statistical analysis of the extracted data from selected literatures. The percentage of organic N mineralized (Nm) and apparent mineral N release (Nr) from BBFs were both calculated as indicators for the N mineralization potential within 14, 45 and 90 days of incubation. A semi-quantitative analysis was conducted to identify the effects of different BBFs properties on plant N uptake efficiency (NUE) and N fertilizer replacement value (NFRV) in the first year of BBF application.

The BBFs such as digestates and liquid slurry fractions that recorded in this study tended to achieve high mineral N release in a short-term mainly because of their high share $NH_4$-N of total N content. The statistical results also indicated that $NH_4$-N/total N content of BBFs had a significantly positive effect on short-term mineral N release. On the other hand, C/N ratio showed a negative effect on N mineralization potential, which explained partially why BBFs with high C/N ratio such as composts and straws resulted in lower or even negative net N mineralization in a short-term of incubation. Other parameters such as organic N, total N and $NH_4$-N content were also found to significantly influence N mineralization potential in different time periods. The BBF properties on total N, $NH_4$-N, $NH_4$-N/total N ratio and C/N ratio had significant effects on NUE and NFRV. The $NH_4$-N/total N ratio had the strongest positive linear correlation with NUE, whereas the C/N ratio had the best fit with NFRV in a negative logarithmic regression model.

Overall, it is suggested that the BBF with higher $NH_4$-N/total N ratio and lower C/N ratio is likely to achieve higher short-term N mineralization and maintain a high NFRV for the first-year application.

**Key words:** Bio-based fertilizers; short-term N mineralization; N uptake efficiency; N fertilizer replacement value

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# List of Abbreviations

| | |
|---|---|
| AE | Agronomic Efficiency |
| ANOVA | Analysis of variance |
| ANR | Net apparent N recovery |
| BBF(s) | Bio-based fertilizer(s) |
| C | Carbon |
| DM | Dry matter |
| K | Potassium |
| LR | Linear regression |
| MIT | Mineralization and immobilization turnover |
| MLR | Multiple linear regression |
| MOW | Municipal organic waste |
| N | Nitrogen |
| Nm | Organic N mineralized |
| Nr | Apparent mineral N release |
| NFRV | Nitrogen fertilizer replacement value |
| NUE | Nitrogen uptake efficiency |
| P | Phosphorus |
| PAN | Plant available N |
| RC | Regression coefficient |
| SLR | Systematic literature review |
| WWTP | wastewater treatment plants |

# 1. Introduction

## 1.1 Background Information

Nitrogen (N) is the essential element in the formation of proteins and nucleic acids for every living form on earth. As one of the most crucial and limiting nutrients for crop production, N has been applying to the soil through fertilizers in different forms extensively (Rütting, Aronsson and Delin, 2018; Omara, Aula and Raun, 2019). The common fertilizer nowadays is synthetic mineral N fertilizer produced by the energy-intensive Haber-Busch method, which consumes a significant amount of non-renewable fossil fuels. Even though the emergence of synthetic N fertilizer has greatly increased global crop productivity, it could also bring serious environmental impacts when it is overused with poor fertilizer application strategy. More specifically, single long-term application of mineral fertilizers would disturb soil structure, resulting in the decline of soil organic matter (Fu et al., 2014; Pei et al., 2015) and may easily trigger environmental problems such as soil degradation and surface water eutrophication through nutrient leaching, etc. To move forward to a sustainable agriculture with minimum negative fertilizer impacts, there are needs to reduce synthetic mineral N fertilizer consumptions with better management strategies and replace synthetic N fertilizers partially or entirely by potential organic-based alternatives.

Animal manures are typical farm-based alternatives that have been applied extensively for centuries. Nowadays, organic by-products from municipal sources such as food wastes and sewage sludge from wastewater treatment plants (WWTP) are potential raw materials for N bio-based fertilizers (BBFs) production. Meanwhile, industrial organic by-products from slaughterhouses, fishery and food & beverage industries are also considered as possible organic sources. Some of these organic materials can be applied to the soil directly after collection, whereas others may need to go through various processes such as mechanic separating, sanitizing, composting or digesting, so that they are qualified as organic N fertilizer for field application. Many BBFs are rich in N and also contain other macronutrients such as phosphorus(P) and potassium(K). It is possible to utilize the bio-based materials as fertilizers and gradually reduce the consumption of mineral N fertilizer in intensive agricultural activities, or even fully replace mineral N fertilizer application such as in organic farming. Plaza-Bonilla *et al.* (2017) found that the grain yields in the first year from N-rich BBFs (such as pig slurry) treatment are similar to or can be even greater than mineral N fertilizers treatment when BBFs have been applied appropriately. Besides, if the release of inorganic N from BBFs synchronizes with the demand of crop growth along with the growing period, the potential leaching can be reduced and fertilizer efficiency would be improved (Adegbidi and Briggs, 2003).

BBFs can not only offer nutrients for crop growth but also help achieve good soil quality and soil fertility. Soil quality is the broad reflection of soil functions on regulating soil water retention, sustaining the life of flora and fauna in soil, filtering potential pollutants, cycling nutrients and supporting physical structures (USDA, 2020). Soil fertility is narrowly termed as the ability of soil to provide a sufficient, balanced, toxic-free amount of nutrients and maintain a suitable environment for crop growth. Soil fertility is controlled by original soil textures, soil organic matter (SOM) concentration and plant nutrient level. Evidence showed that organic manure and composts could significantly increase total soil carbon (C) and N level (Mylavarapu and Zinati, 2009). Therefore, with appropriate fertilizer practice, BBF application could help soil regain good quality by restoring the degraded soil, improving soil structure and water retention, stimulating microbial activities, keeping the nutrient cycle( e.g. C and N turnover) in dynamic movement and maintaining short/long-term soil fertility (Fließbach *et al.*, 2007; Abbasi and Khizar, 2012; de Groot and Bogdanski, 2013). Once the chemical and physical properties of soil are improved by BBFs, the crop yield and N use efficiency could be increased subsequently.

Bio-based organic materials that have lost their value (also referred to as wastes) in the previous production lines are collected and flowed into agricultural activities as plant & soil nutrient sources (mainly as N sources for crops in this study), which is the reflection of sustainable material flow and circular economy (Vaneeckhaute *et al.*, 2013; Cobo, Dominguez-Ramos and Irabien, 2018; Diacono *et al.*, 2019). In addition, if a certain amount of mineral N fertilizers can be replaced by BBFs, the energy and non-renewable resources that are consumed during mineral N fertilizer production can be saved, and the negative environmental impacts during synthetic mineral N production are indirectly mitigated (Cobo, Dominguez-Ramos and Irabien, 2018).

Even though BBFs can bring various benefits to sustainable agriculture practices, the application of BBF should be more careful to avoid potential negative impacts. As mentioned before, the various nutrient composition of BBF is a great asset that makes them differ from mineral N fertilizers. However, it also causes difficulties in setting the optimal application rates. For instance, Sigurnjak *et al.* (2017b) argued that when fertilizers with both high P and N content (such as pig slurry and digestates) are applied according to crop N demand, the P input rate would exceed the maximum local allowance of 80kg/ ha. As a result, application rates of BBFs had to be reduced according to the P threshold, which caused the single application of organic fertilizers not being able to offer sufficient N for crop growth. Keskinen *et al.* (2020) also stated that when application rates of poultry manure are based on crop N demands, there is a chance of excess P input. BBFs such as farmyard manure or compost release available N slowly, thus may not fully replace the use of mineral N fertilizer under limited application rate in a short period. Besides, when BBFs are applied regularly or excessively to achieve high yield, nutrient leaching (especially N and P)

would possibly occur and bring environmental problems(Gutser *et al.*, 2005; Pu *et al.*, 2012; Keskinen *et al.*, 2020).

The main challenges of BBFs utilization are how to synchronize the crop N demand with N release pattern of different BBFs and meanwhile to avoid N losses as much as possible(Benke *et al.*, 2017). A better estimation on mineral N release potential of different BBFs is required before application in order to utilize BBFs in a resource-efficient way and minimize the impacts of N surplus to the environment (Svanbäck *et al.*, 2019).

## 1.2 Research Objectives

In this paper, the main focus will be on the ability of BBFs offering plant available N, compared to the synthetic mineral N fertilizers in a short-term period. Therefore, the objective of this study is to answer the following related research questions:

(1) Which factors influence the bioavailability of N in soil after BBF application?

(2) Which properties are related to N mineralization on releasing available N in different types of BBFs?

(3) Is the plant N uptake efficiency (NUE) or nitrogen fertilizer replacement value (NFRV) of BBFs related to the same factors and properties that influence N mineralization potential?

# 2. Literature review

## 2.1 Bio-based fertilizers (BBFs)

According to the objectives, the performance of BBFs as N fertilizers will be the main focus. Therefore, in this study, BBFs are defined as fertilizers that are produced by natural-based substances or recycled materials to provide N for crop growth. BBFs can be broadly categorized as 1) untreated raw organic residues such as animal manures, plant-based residues or sludge from WWTP industry; and 2) processed compost, digestates, mineral concentrates or commercialized organic/inorganic pellets.

### 2.1.1 BBF sources

**Agricultural wastes**

Agricultural wastes are a group of organic substances derived from agricultural production, which includes livestock manure, plant-based green manure and forest residues (Chojnacka, Moustakas and Witek-Krowiak, 2020). Slurry is a well-used term to describe the mixture of feces and urine with bedding materials (wood shaving, sawdust and straw) and water, which contains a high proportion of $NH_4$-N(Pain and Menzi, 2003). Poultry litter or broiler litter is a mixture of poultry manure, bedding materials (e.g., straw, sawdust paper and wood shavings), spilled feed and feathers. It is created during the operation of poultry farms, which consists of 60–70% ammonia or uric acid (Huang *et al.*, 2015). Deep litters refer to the mixture of animal excreta and a larger proportion of bedding materials. Solid manures such as farmyard manure or cattle manure are characterized as high DM content, high C/N ratio and low $NH_4$-N content that release inorganic N slowly and may fail to provide enough N in short growing period when a limited amount of manures are applied on field(Gutser *et al.*, 2005; Sørensen and Thomsen, 2005; Cavalli *et al.*, 2016; Mishra *et al.*, 2019).

Crop residues are unprofitable and inedible parts of crops such as leaves and straws, which are either left as mulch or collected and then re-incorporated into the field as green manures (Jat, Bijay-Singh and Gerard, 2014). Mulching is an easy way to manage the crop residues and has benefits on conserving soil moisture and structure, maintaining stable soil temperature and improving soil organic matters(Muimba-Kankolongo, 2018). However, even though part of N in mulch can be taken up by subsequent crops (Dahlin, Stenberg and Marstorp, 2011), the accumulation of high amounts of N in field may occur (Brozyna *et al.*, 2013; Carter *et al.*, 2014). This may increase the risk of N loss through $N_2O$ and $NH_3$ emissions (de Ruijter, Huijsmans and Rutgers, 2010; Dahlin, Stenberg and Marstorp, 2011; Nadeem *et al.*, 2012). Therefore, the second

option could be taken to remove green manure cuts from the field and to store or reprocess the biomass as ‘‘mobile green manure’’ for further upgrading treatments such as composting or anaerobic digestion (Nygaard Sorensen and Thorup-Kristensen, 2011; Sørensen *et al.*, 2013).

**Municipal organic wastes**

Biodegradable garden and park waste, food and kitchen waste from households, restaurants and municipalities are defined as municipal organic wastes (MOW). The bio-wastes from municipal sectors have a high potential for nutrient recovery. A study by (Zabaleta and Rodic, 2015) found that N recovery from MOW was significantly low in the city of Vitoria-Gasteiz ( Spain), mainly because of the inefficiency of local municipal organic waste management system. In addition to the small amount of urban bio-waste that is recycled, the composition and quality of municipal organic wastes vary greatly between regions with different dietary habits and organic waste collection policies. Due to the heterogeneity of municipal organic wastes, it is unable to apply them directly to soil as fertilizers unless being specially treated to achieve stability and meet N fertilizer requirements. Another type of organic waste from the municipal sanitation stream is human urine and feces, which have also been considered as a strong source to provide nutrients for agricultural activities (Andersson, 2015).

**Sewage sludge**

Sewage sludge and biosolids are organic by-products from water cleaning up process in municipal and industrial wastewater treatment. The term ‘biosolid’ has come to be used as the stabilized and sanitized organic solid residuals from sewage sludge (Rigby *et al.*, 2016). Sewage sludge has positive effects on crop production and soil fertility because it normally contains a wide variety of elements, especially on N and P (Suhadolc *et al.*, 2010). However, sewage sludge may also carry unwanted components such as pathogens and heavy metals. Kominko, Gorazda and Wzorek (2019) investigated several sewage sludges from different resources in Poland and discovered that all the sewage sludge samples contain at least one heavy metal that the concentration exceeded the legal threshold for fertilizers. It is estimated that intensive and long term sewage sludge input in agriculture may lead to an accumulation of heavy metals and organic contaminations in soil (Kirchmann *et al.*, 2017; Yang *et al.*, 2018).Therefore, the amount of sewage sludge applied as fertilizer should be controlled by its heavy metal concentration in order to maintain the heavy metal input lower than the limited values (EU Sludge Directive, European Commission, 1986). Advance treatment methods such as solar drying, thermal treatment, chemical extraction and biological treatments could increase the fertilizer potential of sewage sludge by reducing

phototoxicity and heavy metal contents, stabilizing sludge component and improving N & P recovery (Li, Gan and Hu, 2016; Kominko, Gorazda and Wzorek, 2019; An-nori *et al.*, 2020).

**Industrial organic wastes**

Food processing wastes represent the food materials that are discarded during the entire supply chain (Du *et al.*, 2018). For instance, blood, bone and meat meals are common commercial organic fertilizers that originate from slaughterhouse or animal products industry. Organic residues from bioenergy plants such as rapeseed meal mainly contain a higher amount of easily degradable residues(Galvez *et al.*, 2012). Waste keratin materials such as chicken feathers and untreated feather meal from poultry industry include up to 15 % of N on dry matter (DM) basis (Nurdiawati *et al.*, 2018). However, raw feathers are unable to be used directly as N fertilizers because of the rigid, hardly degradable structure(Onifade *et al.*, 1998). With specific treatments such as hydrothermal processing methods(Nurdiawati *et al.*, 2019), feathers can be transformed into amino acids and short-chain peptides as the primary source of N for crop growth(Endres and Mercier, 2003). The fishing and fish processing industry generates about 30%-70% fish wastes per ton fish products (Ahuja *et al.*, 2020). Those fish wastes rich in N and P are potential raw organic materials as fertilizers (Radziemska *et al.*, 2019). Food & beverage processing factory generates various organic residues along with the production line: for instance, beer vinasse from beer brewing industry (Wang *et al.*, 2017), grape marc from wine production(Flavel *et al.*, 2005), spent coffee grounds (Stylianou *et al.*, 2018) and spent mushroom substrate(Zmora-Nahum, Hadar and Chen, 2007). It is needed to notice that most raw organic residues are in general unable to be applied directly as fertilizers because of their size and complicated compositions. Special treatments are mainly required to minimize negative effects such as undesirable pH value and heavy metals, and to increase the share of mineral N and degradability of organic N in different organic materials.

### 2.1.2 Treatment methods of raw materials for BBF production

As mentioned in section 2.1.1, some organic residues from either agricultural or municipal sources should be further treated to reduce the negative effects on plants and soil and increase their N fertilizer potential. Different treatments such as biological, physical, chemical or combined methods are chosen in the fertilizer upgrading process so that specific organic residues can be adjusted into more stable and degradable N fertilizer products.

**Biological treatment**

- **Composting**

Composting is defined as a biological treatment in which aerobic thermophilic and mesophilic indigenous microorganisms consume organic matter as substrates or nutrient sources (de Bertoldi, Vallini and Pera, 1983). The N forms in composts mainly depend on the composition of raw materials and the specific composting processes such as temperature, selection of adding materials and duration. The composting process can lower the volume and weight of the original biowaste, reduce pathogens and weed seeds, and transform biowastes into stable high-quality soil amendments or slow-released organic N fertilizers (Bernal, Alburquerque and Moral, 2009). Due to the heterogeneous of municipal organic wastes (MOW), composting is mostly required in order to transfer MOW into organic fertilizers or amendments with high organic matter content and low bulk density (Soumaré, Tack and Verloo, 2003). An incubation study by Gale *et al.* (2006) indicated that composts are more likely to be stable and predictable on supplying available N regardless of diverse raw materials. However, nutrient loss is inevitable during composting process and it was reported that the N loss is ranged from 16% to 76% during composting of different materials(Barrington *et al.*, 2002). Grigatti *et al.* (2011) stated that the solid fraction of biogas sludge after composting showed a significant mass loss (25.3%) due to the intense organic matter decomposition. Moreover, the longer the composting process, the more recalcitrant organic N remains in the composts, thus less net inorganic N will be released into soil in a short period from the mature composts compared to fresh composts(Hartz and Johnstone, 2006).

- **Anaerobic digestion**

Another common biological treatment is anaerobic digestion, which converts organic residues into renewable energy (biogas) and digestates. During anaerobic digestion, a large proportion of C is converted to methane ($CH_4$), and in the meantime, organic N compounds are decomposed into mineral forms or easily degradable materials. Therefore, digestates are usually rich in bioavailable nutrients and easily biodegradable organic matters, which especially increased $NH_4$-N share compare to the untreated raw materials (De Notaris *et al.*, 2018). In other words, anaerobic digestion raises potential N availability of digestates and reduce the C to organic N ratio, which makes the digestates competitive to mineral N fertilizers for short-term fertilizer application (Gutser *et al.*, 2005; Cavalli *et al.*, 2016). The composition of digestate depends on the nature of the source material and the anaerobic digestion process (Alburquerque, de la Fuente and Bernal, 2012). In general, solid fractions of digestate contain a significantly higher organic matter and P content, whereas liquid fractions tend to have higher concentrations of $NH_4$-N. Therefore, solid fractions of digestates have a greater potential as soil amendments, whereas the liquid fraction has greater potential as N fertilizer (Tambone and Adani, 2017).

**Physical and chemical treatment**

Physical methods such as solid-liquid separation, pelletizing, granulating and thermal heating are commonly used techniques in organic fertilizer upgrading treatments. For instance, when slurry is separated into liquid and solid fractions, the former is rich in soluble N content and contains lower dry matters (DM), which can be incorporated into the soil as efficient N fertilizers. On the other hand, solid fractions with rich organic matters are able to transport at low costs per unit and be further processed (e.g., composting pelletizing or granulating) into economically transportable, sanitized organic fertilizer products (Fangueiro *et al.*, 2014). Besides, manures or other solid organic wastes are often pelletized into certain sizes, mostly aim to produce marketable organic fertilizers(Rao *et al.*, 2007). Thermal treatments of bio-based wastes have the advantages of high temperature for the elimination of pathogens and microorganisms (Cantrell *et al.*, 2012) which is therefore normally applied for sewage sludge as pre-treatments. However, Paneque *et al.* (2017) pointed out that thermochemical conversion is primarily targeted for producing P-rich fertilizers since majority of the feedstock N may be lost through gas emission. (Rouch *et al.*, 2011) also mentioned the air-drying treatments significantly reduced N concentrations in the dry materials compared to fresh raw material. Additionally, hydrothermal treatment (HTT) combines heat and water as media for wastes treatment, pre-treatment for anaerobic digestion(Bougrier, Delgenès and Carrère, 2008)and nutrient recovery (Nurdiawati *et al.*, 2018). This has been proven to be a potential method to convert protein-rich waste such as chicken feathers, which have poor biological degradability, into a liquid product with higher fertilizer N availability(Nurdiawati *et al.*, 2019). Other specific treatments such as ultrasonic treatment(sonification) and alkalinization are pre-treatment of anaerobic digestion to increase the degradability of organic matters (Naddeo *et al.*, 2009).

Several advanced treatments are even capable of extracting inorganic N intensively, such as membrane filtration(Velthof, 2015), reverse osmosis treatments, ion exchange and stripping technologies (Limoli, Langone and Andreottola, 2016). Those mentioned high-tech separations are able to recover $NH_4$-N from animal manures or liquid digestates to concentrate mineral forms (about 90% are $NH_4$-N) (Zarebska *et al.*, 2015; Sigurnjak *et al.*, 2019). The so-called mineral concentrate N fertilizers have almost the same performance on N supplements as conventional mineral N fertilizers (Vaneeckhaute *et al.*, 2013; Schröder *et al.*, 2014; Sigurnjak *et al.*, 2019). While it should be noticed that the mineral concentrates normally cannot fully replace the conventional fertilizer, which could be because 0-10% of that concentrates still remains in organic N forms(Schröder *et al.*, 2014).

## 2.2 N dynamics in agricultural system

Soil environment changes subsequently along with the input of BBFs. Understanding the dynamic soil N cycle is important to further estimate the quality of BBFs as alternatives for mineral fertilizers(Calderón, McCarty and Reeves, 2005). There are several crucial N turnover processes happening simultaneously in soil, which include depolymerization, N mineralization /immobilization, N nitrification/denitrification, plant N uptake, N leaching and gaseous N emission. The paradigm of soil N transformations with BBF input is shown in Figure 2.1.

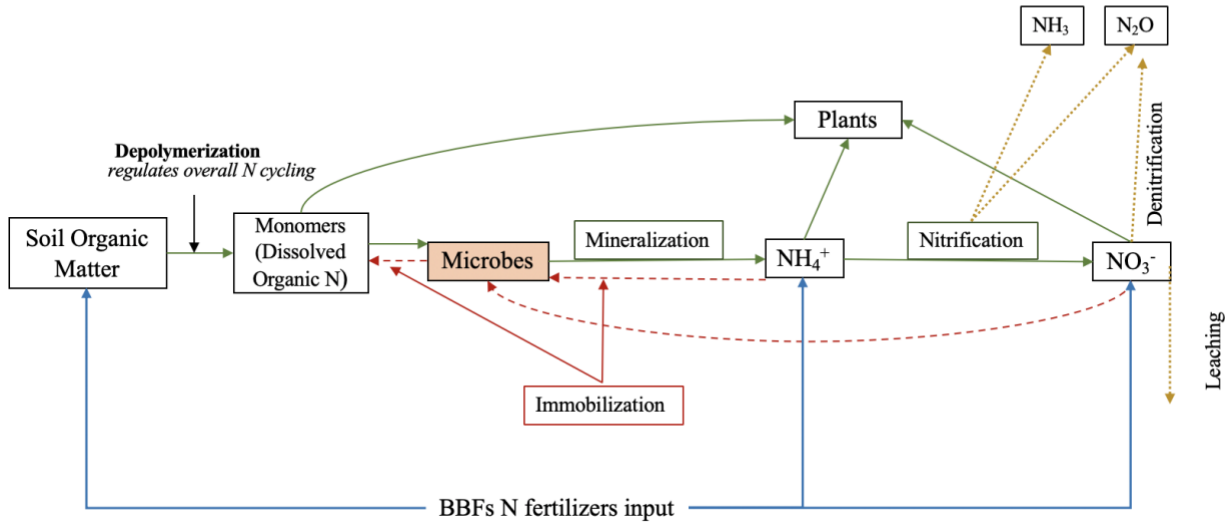



**Figure 2.1 Soil N cycle developed in the late 1990s. Modified from (Chapin, Matson and Vitousek, 2011)**

Generally speaking, plants can only take up inorganic N ($NH_4^+$ and $NO_3^-$) directly in soil. In an N-limiting environment, plants can also assimilate dissolved organic low-molecular N compounds such as amino acids (Schimel and Bennett, 2004; Jones *et al.*, 2005; Huygens *et al.*, 2008), but this may not be the case in agricultural practices where N is supplemented accordingly. The rest of organic N which plants cannot immediately utilize would be transformed into $NH_4^+$ gradually with the help of soil microorganisms, which refers to the mineralization process(Jarvis *et al.*, 1996; Jensen, 2013). $NH_4$-N can be further transformed into $NO_3$-N with the help of nitrify bacteria in aerobic condition, namely the nitrification process. The nitrogen forms that can be taken up by soil microbes and plant species are defined as bioavailable N (Schimel and Bennett, 2004). Plant available N (PAN) is explicitly termed as the sum of inorganic N ($NH_4^+$ and $NO_3^-$) and organic N that can be mineralized gradually in soil during the growing period. Potential mineralizable N

indicates the amount of organic N which can be potentially mineralized from the BBFs within a period and under certain conditions. The mineralization patterns of BBFs vary enormously and are influenced by many internal and external factors. Therefore, it is important to estimate BBF organic N mineralization rate in order to apply BBFs at optimal rate and time.

### 2.2.1 N mineralization and immobilization

Mineralization happens when soil microorganisms utilize organic N compounds as an energy source and release $NH_4$-N to soil. Nitrogen immobilization is the opposite process of mineralization when soil microbials assimilate $NH_4^+$, $NO_3^-$ and amino acids into organic N constituents. These two processes are intimately connected and always happen simultaneously (Jarvis *et al.*, 1996). Therefore, net N mineralization/immobilization is termed as the difference between gross mineralization and immobilization. The continuing transformation process between organic and inorganic N in soil is also known as Mineralization-Immobilization-Turnover (MIT). The variability of physical and chemical characters of BBFs may cause different MIT patterns in soil after fertilizer application. For instance, Abbasi and Khaliq (2016) discovered that the mineralization rate of poultry manure (C/N ratio at 13) increased progressively from the start of incubation and 48% of total N are released as inorganic forms at the end of incubation period (140 days). However, the wheat straw residues (C/N ratio at 46) showed N immobilization in the first half incubation period, net N mineralization only started from day 84 to 140, and only 19.4% of total N remained in mineral forms in the end of incubation. Cavalli *et al.* (2016) found that high C/N ratios and high proportion of volatile fatty acids, cellulose and hemicellulose in the solid fraction of digestates and untreated cattle slurry stimulated net N immobilization(Sørensen, 1998; Morvan and Nicolardot, 2009).

### 2.2.2 N nitrification and denitrification

Nitrification is another microbial process in soil when nitrifiers consume $NH_4^+$ and release $NO_3^-$ and $NO_2^-$ under aerobic conditions(Jarvis *et al.*, 1996). Abbasi *et al.* (2007) observed that during poultry manure soil incubation, nitrification started after $NH_4$-N accumulated up to 40 days, and $NO_3^-$ increased consistently from day 55 to the late stage of incubation. This pattern of nitrification was similar to other studies that the mineralized $NH_4$-N nitrified quickly in favorable aerobic conditions(De Neve *et al.*, 2004; Grigatti *et al.*, 2011; Galvez *et al.*, 2012; Sigurnjak, *et al.*, 2017b). Denitrification is the process consuming $NO_3^-$ under anaerobic condition coupling with nitrification in soil and manure storage systems(Jensen and Sommer, 2013), which is a potential

pathway of $NO_3$-N loss from soils after BBF application. In addition, during N mineralization, the oxygen in soil is consumed rapidly by microbes and thereafter may cause denitrification with $N_2O$ emission (Aulakh, Khera and Doran, 2000; Hadas *et al.*, 2004; Calderón, McCarty and Reeves, 2005). This explains partly the significant N loss accompanied by the shortage of plant-available nitrogen in the soil (Weber et al., 2014). The $N_2O$ emission may be also triggered by the nitrification process. Galvez *et al.* (2012) observed the $N_2O$ emission even under aerobic condition with moderate soil water holding capacity (40%), which could be because the slow organic N mineralization from BBFs failed to offer enough $NH_4$-N and thus stimulated the $N_2O$ formation through nitrification (Wrage *et al.*, 2001).

### 2.2.3 Influencing factors on BBFs N bioavailability

Factors that may affect BBFs N bioavailability have been discussed widely by previous publications, which can be roughly categorized as 1) inherent BBFs properties on mineral N concentration and organic N degradability and 2) external conditions during and after fertilizer application such as soil properties and application methods.

**BBFs properties**

BBFs physical and chemical properties are influenced by the source and type of raw organic material, storage time of organic residues and treatment methods (Zmora-Nahum, Hadar and Chen, 2007). Manure composition varies in relation to animal type, age, diet and management system(Chadwick *et al.*, 2000; Sørensen and Fernández, 2003; Powell *et al.*, 2006). For instance, untreated cattle slurries contain mostly half of mineral N ranging from 49% to 52% (Schröder, Uenk and Hilhorst, 2007) and the $NH_4$-N/total N ratio of untreated pig slurry is around 31% (Fangueiro *et al.*, 2014). The mix of different bedding materials and storage conditions also influence mineral N concentration and the degradability of organic N in the final outcomes as fertilizers(Beegle *et al.*, 2008). Chadwick *et al.* (2000) pointed out that N loss is inevitable during BBF upgrading processes such as drying, pelletizing and composting. Municipal wastes after composting reduced half of the organic C compared to the untreated one, and there was a moderate decrease on total N after MOW composting which is likely due to $NH_3$ emission(Busby, Allen Torbert and Gebhart, 2007).

Subsequently, BBFs properties on mineral N concentration and organic N degradability regulate the N mineralization/immobilization process in soil after fertilizer application (Van Kessel and Reeves, 2002; Cabrera, Kissel and Vigil, 2005; Flavel and Murphy, 2006). Moe *et al.* (2020) found

that about 73% organic N was mineralized in poultry manure after 12 weeks of incubation; in contrast, only 27% of the organic N in cattle manure was mineralized within the same incubation time. This is likely because poultry manures contained higher amount of easily degradable organic N such as proteins and uric acid than cattle manures (Bozkurt *et al.*, 2006). Helgason *et al.*, (2007) also discovered that less than 5% organic N was degraded in cattle manure composts over 425 days, suggesting that the composts contain large amount of hardly degradable organic matters such as cellulose or lignin. Some types of BBFs such as slurries and digestates are higher in initial NH4-N content, which may offer sufficient N in the initial stage (Cavalli *et al.*, 2017). Therefore, the $NH_4$-N/ total N ratio to some extent indicates the ability of BBFs offer mineral N in short period.

The amount of C in soil is the limiting factor for soil microbe growth (Miller and Geisseler, 2018). When BBFs with both high C and N content are applied, soil microbe communities would increase significantly, and the N mineralization process might be also improved. However, when C content is high, but N content is not enough to support the need of growing microbes (high C/N ratio), immobilization may occur. Therefore, the C to N ratio should be a good reflection on BBFs N mineralization potential. Chadwick *et al.* (2000) suggested that manure C/N ratios greater than 15 may cause net N immobilization in the initial stage. However, C/N ratio of BBFs did not always well predict the behavior of BBFs on N mineralization after applied to soil (Flavel and Murphy, 2006). When C/N ratio is not enough to reflect plant available N potential of BBFs, other parameters such as water-soluble C & N content (Nahm, 2005) , C/organic N ratio (Galvez *et al.*, 2012) and lignin/total N ratio (Dossa *et al.*, 2009) are taken into consideration.

**Soil condition**

Soil properties such as soil textures, chemical contents, microbial populations, temperature and moisture also influence N mineralization processes after BBF application (Cabrera et al., 2005; Jarvis et al., 1996). Busby, Allen Torbert and Gebhart (2007) discovered that N mineralization rate is significantly higher in the soil with higher indigenous N after the addition of organic fertilizers. Antil *et al.* (2011) also found that after 42 days of incubation with different composts, the mineral N release was higher when soil contained higher initial mineral N.

The soils with high clay content may cause mineral N immobilization thus decrease net N mineralization release (Amlinger *et al.*, 2003; Canali *et al.*, 2011). Besides, the net respiration decreases with the increase of soil clay concentration and the soil microbes become less active. This may be because clay would adsorb organic matters and cause difficulties for soil microbes to consume (Kaiser and Guggenberger, 2003), or the ammonium fixation of clay reduce the exchangeable $NH_4$-N in soil (Delin, 2011; Nieder, Benbi and Scherer, 2011). In addition, N may

be lost through high denitrification potential due to the high percentage of water filled pore space in clayed soil (Linn and Doran, 1984). Soil temperature and moisture level influence soil microbes' activities and then affect N mineralization/immobilization turnover. Net N mineralization rate has was observed to increase with temperature under laboratory incubations (Dalias *et al.*, 2002). Therefore, higher temperature in an optimum range increases the growth of microbes and then stimulates gross N mineralization. However, higher temperature may also stimulate gross immobilization process when organic C is sufficient for microbial activities to consume (Miller and Geisseler, 2018). Cassman and Munns (1980) also found a strong interaction between soil temperature and moisture regime on N mineralization. Soil pH may in some extent influence N mineralization. Galvez *et al.* (2012) observed that N mineralization is in general higher in slight acidic soil than alkaline soil when applied the same BBFs, and it is also proven that soil respiration and enzyme activities are higher. The higher mineral N contents in acidic soil may be also explained by reducing $NH_3$ emission and retain more $NH_4$-N in soil (Catalog, 2008). However, when $NH_4$-N nitrified into $NO_3$-N in the beginning stage (such as digestates with rich $NH_4$-N), the amount of total inorganic N is not likely to be influenced by acidity of soil (Galvez *et al.*, 2012).

## 2.3 Crop N use efficiency and N fertilizer replacement value of BBFs

Agronomic efficiency (AE) and N uptake efficiency (NUE) reflect the plant use efficiency of applied N from different perspectives and are estimated based on different parameters (table 2.1). Agronomic efficiency is calculated as the increase of crop yield (aboveground biomass or grain dry matter) per unit of total N applied compared to yield in unfertilized control group. The change of crop yield along with increasing N fertilization rate is normally shaped as a non-linear positive convex curve, which indicates that AE is influenced by the fertilizer N application rate (Jensen, 2013; De Notaris *et al.*, 2018). N uptake efficiency (NUE), or apparent N recovery (ANR) is calculated as the percentage of total applied N from fertilizers transferred into crops after harvesting. In some pot experiments, the total N content from both shoot and root biomass are measured but most studies on field only considered the total N content from the harvested above ground biomass or even just the profitable parts such as grains from a practical point of view (Jensen, 2013). N fertilizer replacement value (NFRV) indicates the percentage of mineral N fertilizer that can be replaced by BBF application within plant growing season, which specifically reflects the capability of BBFs to replace mineral fertilizers offering plant available N. The NFRV can be calculated based on AE or NUE. It is argued that even though calculating NFRV from AE is more relevant for farmer's concern on marketable yields (Schröder, 2005; Jensen, 2013), the calculated NFRV could be influenced by fertilizer N application rate when the yield response

curve is non-linear (De Notaris *et al.*, 2018). For instance, Hijbeek *et al.* (2018) found that when NFRV was calculated based on the yield response curve of the marketable products, the NFRV of farmyard manure was significant lower at low total N application rate compared to that at high total N rate. If the N fertilizer requirements are based on the NFRV calculated at low N supply, the N supply from BBFs could be overestimated, which may cause excessive N in soil. On the other hand, the crop N uptake is generally linear related to fertilizer N application rate (below the optimal N rate), thus it seems to be more dependable to calculate NFRV based on NUE rather than AE (De Notaris *et al.*, 2018).

**Table 2.1 The equations on agronomic efficiency, net N uptake efficiency and N fertilizer replacement value.**

| Term | Equation | Explanation |
|---|---|---|
| Agronomic Efficiency (AE) | $AE = \frac{Yt - Yc}{total\ N\ applied}$ | Yt: aboveground biomass or grain yields from fertilizers treatment<br>Yc: aboveground biomass or grain yields without fertilizer treatment |
| Net N uptake efficiency (NUE)/ Net apparent N recovery (ANR) | $NUE = 100 * \frac{Nt - Nc}{total\ N\ applied}$ | Nt: aboveground biomass or grain N uptake from fertilizers treatment<br>Nc: aboveground biomass or grain N uptake without fertilizer treatment |
| N fertilizer replacement value (NFRVa) | $NFRVa = 100 * \frac{AEb}{AEm}$ | AEb: AE of BBF<br>AEm: AE of mineral fertilizers |
| N fertilizer replacement value (NFRV) | $NFRV = 100 * \frac{NUEb}{NUEm}$ | NUEb: NUE of BBF<br>NUEm: NUE of mineral fertilizers |

# 3. Methods

Systematic literature review (SLR) is the main approach in this study. The idea of SLR aims to gather relevant results from various scientific literatures that in accordance with the selected criteria to answer specific research questions(Moher *et al.*, 2015). The key steps of SLR in this paper are adapted from Mikolajewicz and Komarova (2019) and Mengist, Soromessa and Legese (2020), which includes (1) defining the study scope; (2) constructing comprehensive searching strategy and conducting broad literature review; (3) screening and selecting literatures from the previous step according to the defined criteria; (4) extracting and collecting relevant data from the final selected literatures; (5) conducting meta-analysis or semi-quantitative analysis; and (6) interpreting key findings and summarizing the results.

According to the research objectives mentioned in section 1.2, a broad literature research was performed firstly based on the summarized Boolean search terms (Table 3.1). Each term was either searched individually or combined with others, for instance: "Nitrogen Mineralization" in "Manure" or "Fertilizer N replacement value" of "Manure". The searching keywords were applied to online databases including Web of Science, Scopus, ScienceDirect, Wiley Online Library and Google Scholar.

**Table 3.1 Boolean research term used in systematic literature review.**

| Category | Boolean research term | Field to which search term was applied |
|---|---|---|
| 1 | N OR Nitrogen | Title |
| 2 | Mineralization | Title |
| 3 | Mineralization rate OR mineralization potential | whole article |
| 4 | Mineralizable N | whole article |
| 5 | Uptake efficiency OR Use efficiency OR recovery efficiency | whole article |
| 6 | Fertilization (N) replacement value | whole article |
| 7 | Mineral N fertilizer equivalent | whole article |
| 8 | Organic fertilizer OR sludge OR manure OR compost OR digestate OR residues | whole article |
| 9 | N availability OR Fertilizer N availability | whole article |
| 10 | NOT mineral fertilizer | Title |

Note: the word spells are either American or British style.

Then, a meta-analysis was conducted to answer research question (1) *Which factors determine the bioavailability of N in soil after BBF application* and (2) *Which properties can reflect N mineralization potential on releasing readily available N in different types of BBFs*. The detailed key steps are described in section 3.1. Subsequently, a semi-quantitative analysis was conducted separately to answer research question (3) *Is the plant N uptake efficiency or nitrogen fertilizer replacement value of BBFs related to the same factors and properties that influence N mineralization potential*. The analytical procedures are presented in section 3.2.

## 3.1 Meta-analysis

### 3.1.1 Data collection and Management

In order to answer the questions on N bioavailability of BBFs (research question 1 and 2), a quantitively meta-analysis was performed to reveal different effects on BBFs N mineralization potential. As part of SLR, meta-analysis aims to combine the experimental results from the publications which meet the established criteria and apply common statistical methods for data analysis. Therefore, the publications included in broad research were selected further based on more detailed criteria, which was shown in Table 3.2.

**Table 3.2 Selection criteria for BBFs property and N mineralization potential.**

| Selection criteria of paper | |
|---|---|
| a | The study is based on aerobic lab incubation experiment and the information on experimental conditions such as temperature and soil properties are available |
| b | The study has at least one treatment without N fertilization as control group |
| c | There is data available on at the required sampling days (14, 45 or/and 90 day) of BBFs N mineralization during the incubation time: $NH_4$-N and $NO_3$-N content are measured timely during incubation day either in table or figure |
| d | Data available on BBF characters of total N, organic N, C/N ratio or more |
| e | Incubation period should be between 14 and 90 days |

BBFs information on Total N, Organic N, $NH_4$-N, $NO_3$-N, Organic C, total P, total K and pH were collected. Experimental data includes BBFs application rate, incubation temperature, soil characters and duration are recorded as well. In agreement with the above criteria, 23 of the studies with 144 observations were included for estimating N mineralization. The information on experimental settings, BBF properties and incubation results (details see supplementary material

1) were collected directly from tables in the literatures. Plot Digitizer 2.6.8 was used if certain data are only available in plot graphs. In order to establish a comparable dataset, BBF properties recorded in fresh weight were converted to the same unit as dry weight (g/kg DM). BBFs included in this study were further categorized into 7 main groups and 24 subgroups according to their original sources, forms and treatment methods (Table 3.3). The categorization of BBFs in this study was also adapted from (Sradnick and Feller, 2020). The animal manures were mainly from local or intensive farms, while sludges were characterized as BBFs from municipal or industrial sources. Green manures were based on fresh or stored plant materials, including the mixtures from several plants in different mixed ratio. There was only one observation on commercial fertilizer, it is therefore located in industrial residue group together with BBFs from food processing and bio-energy industry. Lastly, composts and digestates were classified as organic residues processed via aerobic and anaerobic treatments.

**Table 3.3 BBF categories of main groups (in bold text) and the corresponding subgroups with different number of observations.**

| **Animal manure (39)** | **Slurry (13)** | **Sludge (19)** | **Green manure (10)** |
|---|---|---|---|
| Cattle manure (11) | Pig slurry (11) | Sewage sludge (14) | Untreated green manure (5) |
| Poultry manure (15) | Biogas slurry (2) | Biosolids (5) | Green manure mixture (3) |
| Sheep manure (4) | | | Straw (2) |
| Farmyard manure (2) | | | |
| Pig manure (4) | | | |
| Manure mixture (3) | | | |

| **Industrial residue (6)** | **Composts (45)** | **Digestates (12)** |
|---|---|---|
| Plant residue (5) | Green manure compost (4) | Raw digestates (8) |
| Commercial fertilizer originates from poultry manure (1) | Animal manure compost (26) | Liquid fraction of digestates (2) |
| | Household waste compost (7) | Mineral concentrate of digestates (2) |
| | Sewage sludge compost (11) | |
| | Biogas slurry compost (1) | |

Soil textures were categorized into 10 types according to USDA soil texture triangle. Soil total N, total C content, temperature and moisture level (based on the percentage of water holding capacity) were recorded when available (Table 3.4).

**Table 3.4 The range of different soil properties and experimental conditions within the final dataset.**

| Parameter | Range | Unit |
|---|---|---|
| Total N | 0.08 - 4.8 | g/kg DM |
| Total C | 0.084 - 60.8 | g/kg DM |
| pH | 5.3 - 8.3 | - |
| Temperature | 5 - 30 | $^{o}$C |
| Soil moisture | 40 - 80 | % |
| Soil texture | sandy, clay, clay loam, loamy sand, sandy loam, sandy clay, sandy clay loam, silt loam, silty clay loam, loam | |

Soil properties from the selected studies varied greatly across regions. In order to identify the effects of soil pH, temperature and moisture, these three parameters were grouped separately as categorical variables, which are shown in Table 3.5. The group of soil pH seemed to have similar number of observations in each level. On the other hand, soil moisture contents from different studies varied within relatively optimal values for aerobic biological turnover under control condition. Most experimental designs implemented ideal temperature degree around 25 $^{o}$C, except for the study by Whitmore (2007) and Fontaine *et al.* (2020) which modified temperature at 5 $^{o}$C and 10 $^{o}$C in soil incubation experiments.

**Table 3.5 Categorized groups of soil pH, temperature and moisture in different number of observations.**

| Parameter | Groups | Number of observations |
|---|---|---|
| Soil pH | acidic soil (L, pH ≤6.5) | 56 |
| | neutral soil (M, 6.5<pH≤7.5) | 14 |
| | alkaline soil (H, pH>7.5); | 36 |
| Soil temperature | level 1 (T≤15°C) | 35 |
| | level 2 (15°C<T≤25°C) | 71 |
| | level 3 (T>25°C) | 38 |
| Soil moisture | level 1 (40%~50%) | 30 |
| | level 2 (50%-60%) | 68 |
| | level 3 (>60%) | 26 |

Note: the soil pH group classification was modified from (USDA 1998).

It is known that N mineralization and immobilization are dynamic processes and happen simultaneously, mineral N content in soil may vary in different time points. Therefore, N release in 14 days, 90 days 45 days and were included in this study as early, middle and final stage of

short-term N mineralization, respectively. Once the mineral N content in control and treatment groups at required time point were collected, equations below (Table 3.6) were used to estimate BBFs N mineralization potential from different perspectives:

**Table 3.6 Calculation of BBFs N mineralization potential.**

<table>
<tr><th>Term</th><th>Equation</th><th colspan="2">Explanation</th></tr>
<tr><td>Net mineral N</td><td>Nt=(Nit)BBF-(Nit)control</td><td rowspan="2">Nit: amount of mineral N at sampling time t<br>Ni0: amount of mineral N at sampling time 0</td><td>Net BBF mineral N at time t</td></tr>
<tr><td>Initial mineral N</td><td>N0= (Ni0) BBF - (Ni0) control</td><td>Initial BBF mineral N content</td></tr>
<tr><td>Apparent(net) mineral N release of added N (%)</td><td>$\mathrm{Nr} = 100 * \frac{\mathrm{Nt}}{\text{total N applied}}$</td><td colspan="2">The ability of BBF offer mineral N</td></tr>
<tr><td>Organic N mineralized (%)</td><td>$\mathrm{Nm} = 100 * \frac{\mathrm{Nt} - \mathrm{N0}}{\text{total N applied}}$</td><td colspan="2">BBF release mineral N from organic source as the percentage of applied total N</td></tr>
</table>

The response variable, organic N mineralized (Nm) is calculated as the percentage of mineral N released from BBFs organic N at time t, which to some extent indicates the BBF performance for short term application. However, some types of BBFs such as digestates and liquid slurries contain higher $NH_4$-N and lower organic N, thus Nm may be not able to reflect the performance of BBFs as N fertilizer comprehensively(Case and Jensen, 2019). Therefore, apparent N release (Nr) from both mineral and organic N pools of BBFs was considered as another response variable in the subsequent statistical analysis.

## 3.1.2 Statistical analysis

**Effects of BBF types, soil textures and incubation days on N mineralization**

The effects of BBF types, soil textures and incubation time on Nm and Nr were examined using three-way analysis of variance (ANOVA) tests in the fixed effect model by R software (R development Core Team). Significant test was performed using ANOVA type II or III method when number of observations in different groups were imbalanced. Estimated marginal means (emmeans) were determined for post hoc analysis. Tukey correction was performed to adjust p values with probability of hypothesis rejection at $p < 0.05$ level.

**Effect of BBFs properties**

Multiple linear regression (MLR) was applied to identify the effect of different BBF properties in different incubation time on N mineralization potential. The organic N mineralized (Nm) and

apparent N release (Nr) are the two response variables to represent N mineralization potential, which were analyzed separately (Table 3.7). MLR was conducted in R Studio (version 3.6.2) with function *lm*. The effects of BBF properties on total N, total C, organic N, $NH_4$-N, C/N ratio, C/organic N ratio and $NH_4$-N/total N ratio on Nm and Nr were examined in different sampling time on day 14, 45 and 90. The function *drop1* with "F" tests was used in order to test the significance level of each independent variable, and when the variable is statistically not significant, it will be dropped one by one from the regression model.

**Effects of soil properties**

Multi-way ANOVA was conducted to identify the effect of soil pH, moisture and temperature in different levels as categorical variables on Nm (Table 3.7). ANOVA type III test was used for significance level of different variables when the group sizes are imbalanced. *Emmeans* in R was used combined with Tukey test, which allowed for a pairwise comparison of estimated marginal means of response variable within the different groups of independent variables. The main R codes can be found in Appendix 2.

**Table 3.7 Models performed in statistical analysis.**

| Response variables (Y) | Statistical model | Method |
|---|---|---|
| Nm | Nm~ BBF type * incubation time * soil texture | 3way-ANOVA |
| Nr | Nr~ BBF type * incubation time *soil texture | 3way-ANOVA |
| Nm on day14, 45 or 90 | Nm~ total N + total C+ C/N ratio + $NH_4$-N +… | MLR |
| Nr on day 14 or 45 or 90 | Nr~ total N + total C+ C/N ratio + $NH_4$-N +… | MLR |
| Nm on day14, 45 or 90 | Nm~ soil pH + soil moisture+ soil temperature | 3way-ANOVA |

Note: the statistical models were applied in R software and the "*" mark indicates the interaction effect among independent variables and "+" indicates the interaction effect is not considered.

## 3.2 Semi-quantitative analysis

### 3.2.1 Literature selection and data extraction

In order to answer research question (3) *Is the plant N uptake efficiency (NUE) or nitrogen fertilizer replacement value (NFRV) of BBFs related to the same factors and properties that influence N mineralization potential*, a semi-quantitative analysis was carried out, beginning with re-evaluating the articles from broad literature review and selecting specific literatures based on defined criteria in Table 3.8.

**Table 3.8 Selection criteria for BBFs NUE and NFRV.**

| Selection criteria | |
|---|---|
| a | The study is based on pot or field experiment |
| b | The study has at least one treatment with bio-based fertilizers (BBFs) |
| c | The study has at least one treatment without N fertilization as control group |
| d | There is data available on N uptake or recovery either in form of table or figure |
| e | Only the results from first growing period included |
| f | Results are reported for each treatment on NUE or BBF treatment on NFRV of first year crop growth |

In total, 13 studies were selected with 85 observations included. The information on experimental year, crop types, experimental condition (pot or field experiment), application and harvesting strategies were recorded. Meanwhile, the numeric data such as BBF properties, net N uptake efficiency and N fertilizer replacement value (N uptake basis) were also extracted for data analysis. The general information of each literatures can be found in appendix and the detailed experimental designs were summarized in an Excel spreadsheet (supplementary material). Within the selected studies, most of the experiments were conducted in Europe, except for one in Turkey and Pakistan. It is noticed that the application method of BBFs on field were mainly by soil injection for liquid BBFs and surface incorporation (by hand or spreading machine) for solid BBFs. The testing crops were observed in ryegrass, lettuce, maize, spring barley and winter wheat. Various types of BBFs were included in the dataset, and the BBFs were further categorized based on the classification in former section (Table 3.3). The aboveground biomass was harvested to calculate NUE and NFRV in all field experiments, while shoot and root biomass was collected in some pot experiment studies.

### 3.2.2 Data analysis and results interpretation

In this step, the effect of BBF types and crop species on net N uptake efficiency (NUE) and N fertilizer replacement value (NFRV) were performed using 2-way analysis of variance (ANOVA).

Pairwise comparisons were conducted to identify differences of estimated marginal means among groups. Then, a simple linear regression (LR) was conducted to examine the effect of BBF properties on NUE and NFRV, respectively. In this study, NUE (%) is defined as the net N uptake of plant from fertilizer treatment (N uptake in BBF treatment minus N uptake in unfertilized control treatment) as a percentage of total N applied, and NFRV (%) is calculated by the ratio of NUE in BBF treatment to NUE in mineral N treatment. The recorded BBF properties are total N content (g/kg DM), $NH_4$-N content (g/kg DM), C/N ratio and $NH_4$-N/total N ratio, which are mentioned frequently in literatures as important indicators for NUE and NFRV. Simple linear regression was carried out on R software using *lm* function to examine the effect of each independent variable on NUE and NFRV separately. The variable was log transformed when necessary, correlation coefficient and confidence interval were calculated as well. The statistical models used in data analysis were summarized in Table 3.9.

**Table 3.9 Models performed in statistical analysis in R studio.**

| Response variables (Y) | Statistical model | Method |
|---|---|---|
| NUE or NFRV | Y~ total N content | LR |
| NUE or NFRV | Y~ C/N ratio | LR |
| NUE or NFRV | Y~ $NH_4$-N/total N ratio | LR |
| NUE or NFRV | Y~ $NH_4$-N content | LR |

Apart from data analysis, main findings from the selected studies were examined thoroughly by describing and comparing experimental designs and interpreting the corresponding results from different studies in order to identify the effects of BBF properties and other potential factors on NUE and NFRV in first year application.

# 4. Results

## 4.1 N mineralization potential of BBFs

### 4.1.1 General characteristics of BBFs from the selected literatures

There were 23 individual studies with 144 observations included in this dataset. Figure 4.1 shows the differences of the main characteristic of BBFs among groups. BBFs recorded in this study showed different ranges of variation on different chemical properties. In general, total N varied moderately under 100 N g/kg DM, except for slurry group and several outliers observed from digestate group. The slurry group was observed a great difference on total N content (Figure 4.1 A), ranging from 14.5 to 318.75 N g/kg DM. The extreme variation in slurry group was mainly because the inclusion of only two publications (Grigatti *et al.*, 2011 and Fangueiro *et al.*, 2014), in which the slurries were special treated and separated into liquid and solid fractions: the former generally contained 6 times higher total N content than the latter form. The total C content of BBFs tended to have larger variations in group of animal manures, composts and industrial residues (Figure 4.1 B). As expected, green manures contained high total C content and the values varied within relatively small range with median at 434 g/kg DM. Digestates showed only slightly lower total C content than green manures, with a median value at 420 g/kg DM, which could have been because that the concentration of chemical properties was calculated in dry matter basis. Green manures had greatest variation on C/N ratio among groups. The C/N ratio of the rest BBFs fluctuated slightly around 25 and were mostly lower than 40, with one exception of biosolid with C/N ratio at 95 from Burgos, Madejón and Cabrera (2006). Due to the high total N content, the C/N ratio of digestates were generally lower than other BBF types. The digestates and slurry groups were characterized by their high $NH_4$-N/ total N ratio, for instance, the mean value of digestate was 50% and of slurry 41%. In contrast, the $NH_4$-N/ total N ratio of green manures were almost negligible, with an average value at 1%. In addition, even though the average total N content of green manure (26.5g/kg DM) was similar to animal manures (25.7g/kg DM), the latter contained higher $NH_4$-N content at around 14% of the total N. The minimum, maximum, mean, median and standard deviation of each chemical property for each BBF group were recorded in Appendix 1.

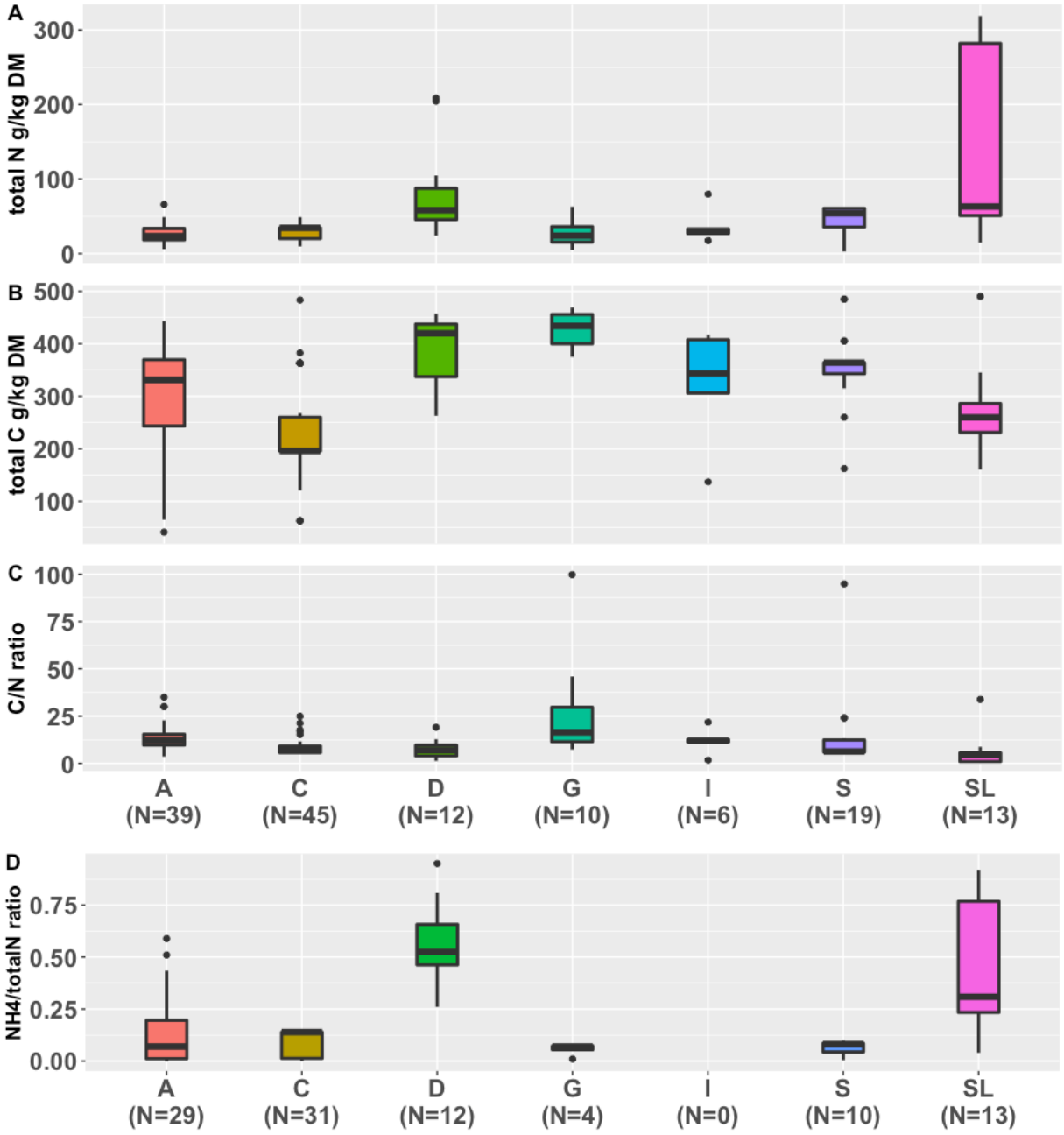



**Figure 4.1 Variation of main BBF properties on (A) total N, (B) total C, (C) C/N ratio and (D) $NH_4$/total N ratio in different BBF categories. The scale of each property differs from each other and $NH_4$-N/total N ratio has less observations due to lack of data. A, animal manures; C, composts; D, digestates; G, green manures; I, industrial residues; S, sludge; SL, slurry.**

### 4.1.2 Overall observations on Nm and Nr from BBF subcategories

From Figure 4.2 to Figure 4.8, the change of Nm and Nr in different subgroups along with incubation time are presented respectively. The results were displayed for each main BBF group separately. Because the experimental results on Nm and Nr were collected from different studies, the number of observations for each subgroup is different, especially for some types of BBFs that were recorded only in one study.

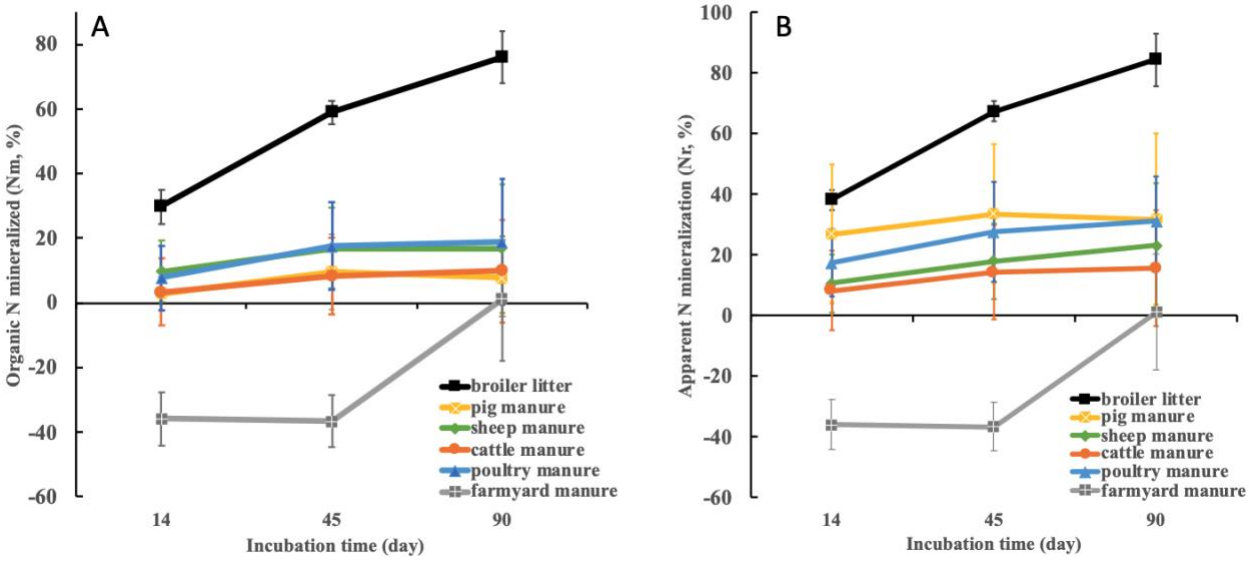



**Figure 4.2 Evolution of Nm(A) and Nr (B) during 90 days of incubation among animal manure. Error bars indicate standard deviation.**

It appears that the group of animal manures had similar Nm and Nr value at each time point, with Nr slightly higher than Nm. Broadly speaking, pig manure, sheep manure, cattle manure and poultry manure have similar performance on organic N mineralization (Nm) at each incubation time. Likewise, apparent N mineralization (Nr) of these four animal manures increased gradually along time, except for the slightly decrease of pig manure from day 45 to day 90. Broiler litter, as the mixture of poultry excreta, spilled feed, feathers and bedding materials, had highest Nm and Nr value at each incubation time (the broiler litter was only from Sistani *et al.* (2008). On the other hand, the farmyard manure from Mohanty *et al.* (2011) had the same and negative value on Nm and Nr, indicating that no initial inorganic N was detected from this farmyard manure.

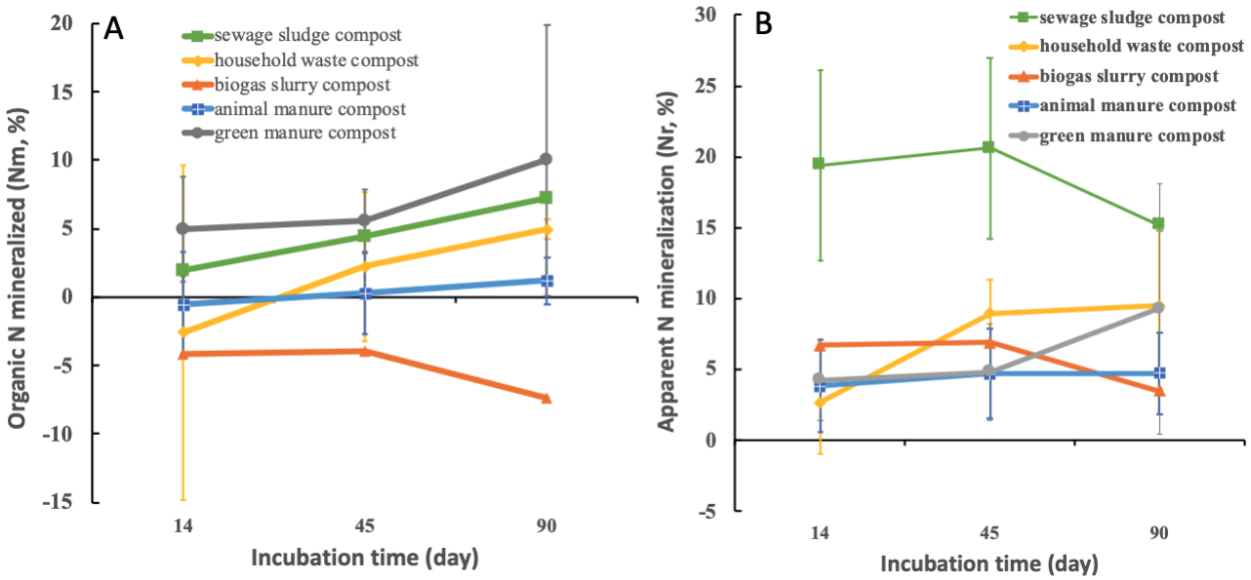



**Figure 4.3 Evolution of Nm(A) and Nr (B) during 90 days of incubation among compost. Error bars indicate standard deviation.**

As expected, organic N mineralized (Nm) from composts in 90 incubation period were generally lower than 10%. Because of the low organic N mineralization rate, composts were often considered as slow-released N fertilizers or soil amendments. The Nm of most compost subgroups tended to increase gradually with time, except for the compost from solid fractions of biogas slurry. The sewage sludge compost achieved higher apparent N mineralization (Nr) at each time point than the rest subgroups, even though a moderate decrease was observed at end stage of incubation (day 90).

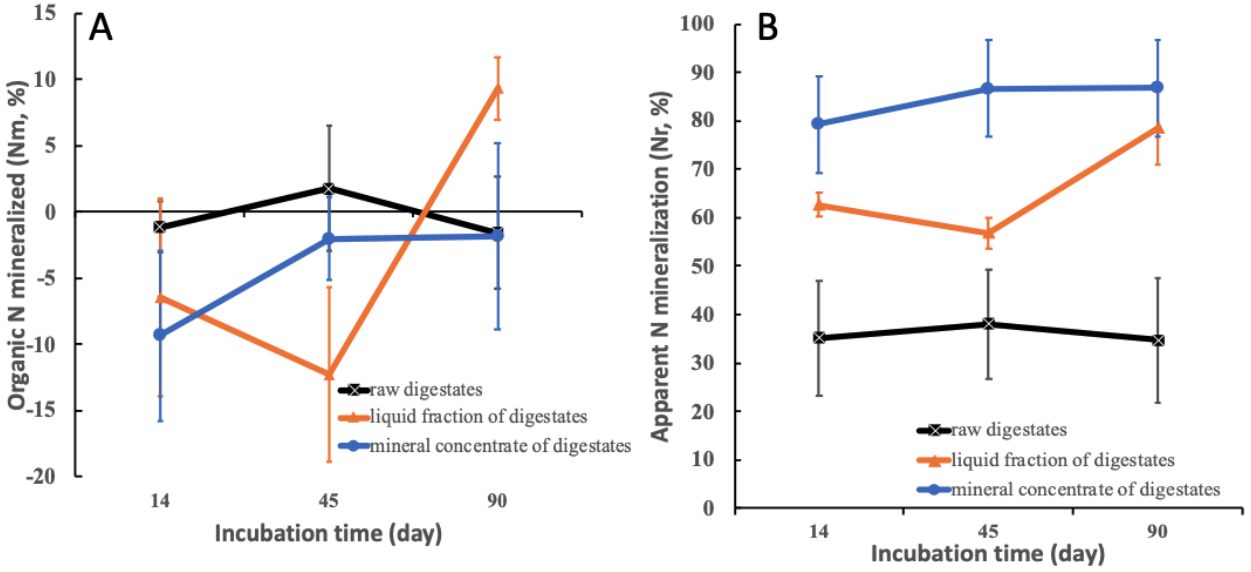



**Figure 4.4 Evolution of Nm(A) and Nr (B) during 90 days of incubation among digestate. Error bars indicate standard deviation.**

As for the digestates, the Nr value was significantly higher than the corresponding Nm, regardless from which sub-categories. The mineral concentrate of digestates achieved highest net mineral N release (Nr) at 79.3%, 86.6% and 86.8% on day 14, 45 and 90, respectively, whereas the Nr value

of raw digestates fluctuated around 35% during the whole 90 days of incubation period. The Nm of raw digestates peaked at day 45 and showed net N mineralization, followed by a decrease under zero. Meanwhile, the Nm of mineral concentrate of digestate increased gradually but failed to reach above zero. The liquid fraction of digestates showed a contradict pattern of change on Nm: net N immobilization was found on day 14 and reached lowest value on day 45. Subsequently, remineralization occurred and increased to positive value of Nm on day 90, with the best fertilizer N potential among subgroups.

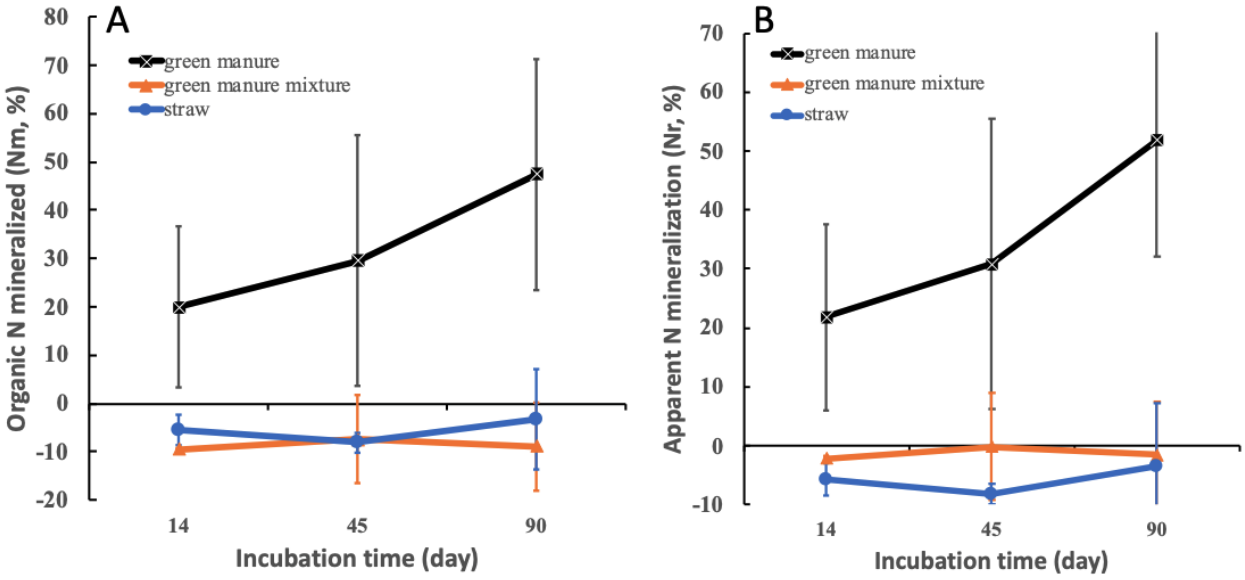



**Figure 4.5 Evolution of Nm(A) and Nr (B) during 90 days of incubation among green manure. Error bars indicate standard deviation.**

Green manure mixture is referred to several plant residues mixed thoroughly in required ratios. It appears that organic N in straw and green manure mixture were hard to decompose, both of which causing mild net N immobilization within 90 days of incubation. On the other hand, organic N of untreated green manures mineralized gradually and reached highest value at day 90.

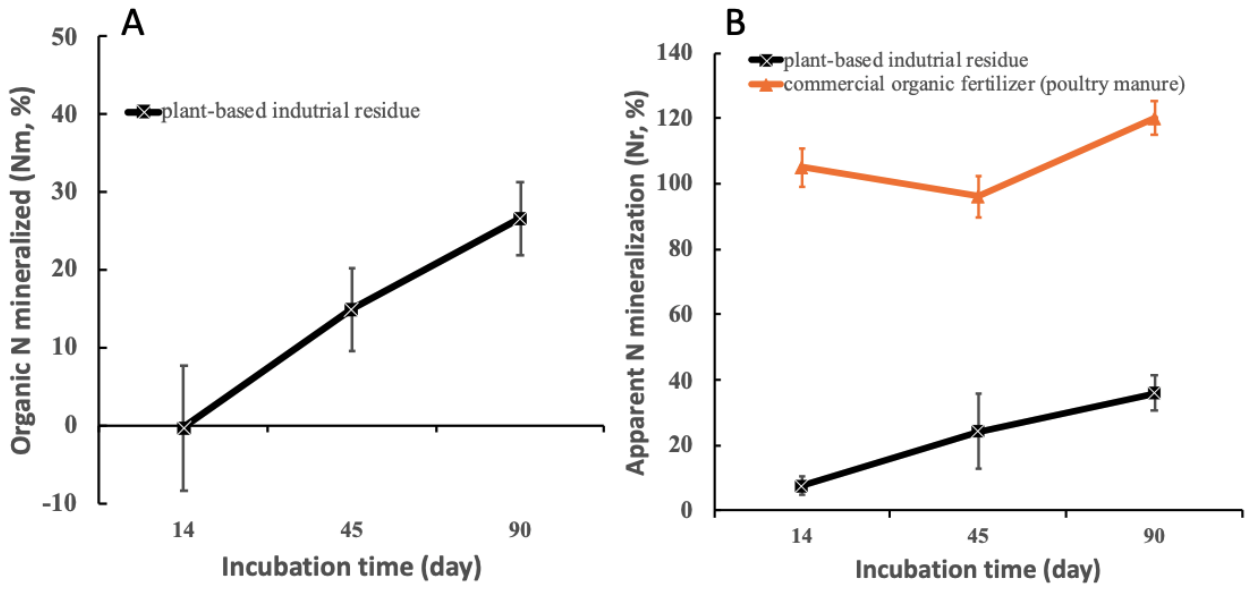



**Figure 4.6 Evolution of Nm(A) and Nr (B) during 90 days of incubation among industrial residues. Error bars indicate standard deviation.**

The plant-based industrial residue had similar performance as the untreated green manures on Nm, where the organic N mineralized moderately along with time. By contrast, the commercial organic fertilizer showed extreme results on mineral N releasing which achieved Nr over 100%, indicating that the application of fertilizer promoted soil indigent organic N mineralization process.

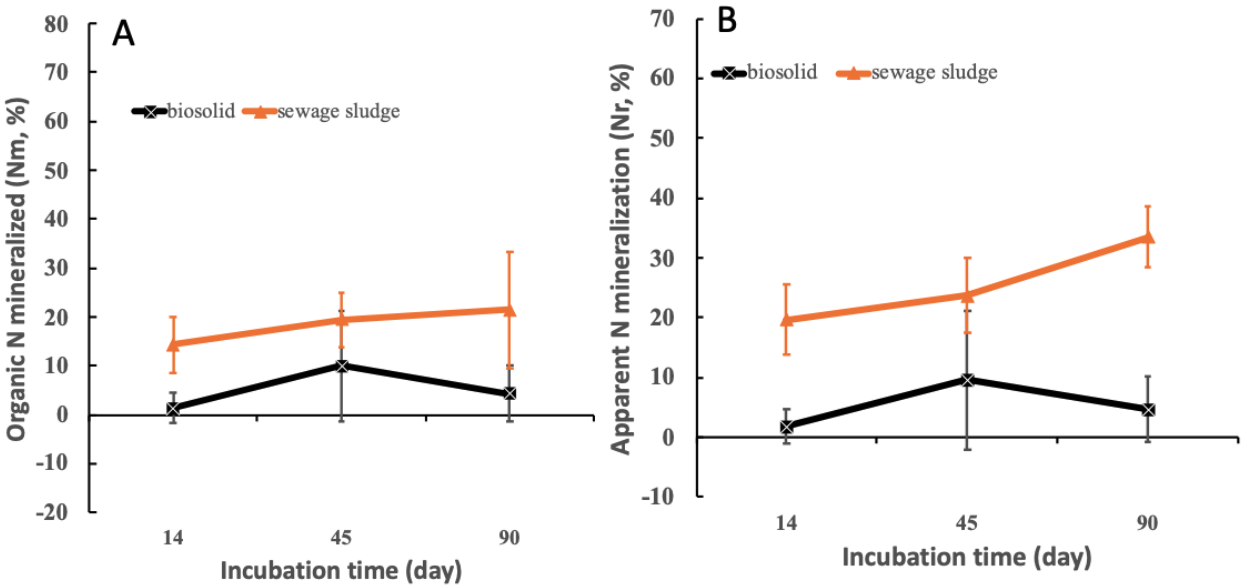



**Figure 4.7 Evolution of Nm(A) and Nr (B) during 90 days of incubation among sewage sludge. Error bars indicate standard deviation.**

Raw sewage sludge had better performance than biosolid both on Nm and Nr at all time period. Besides, the release of mineral N of biosolid tended to peak on day 45, but the N mineralization process of sewage sludge increased gradually and reached highest value at the end of incubation day(day 90).

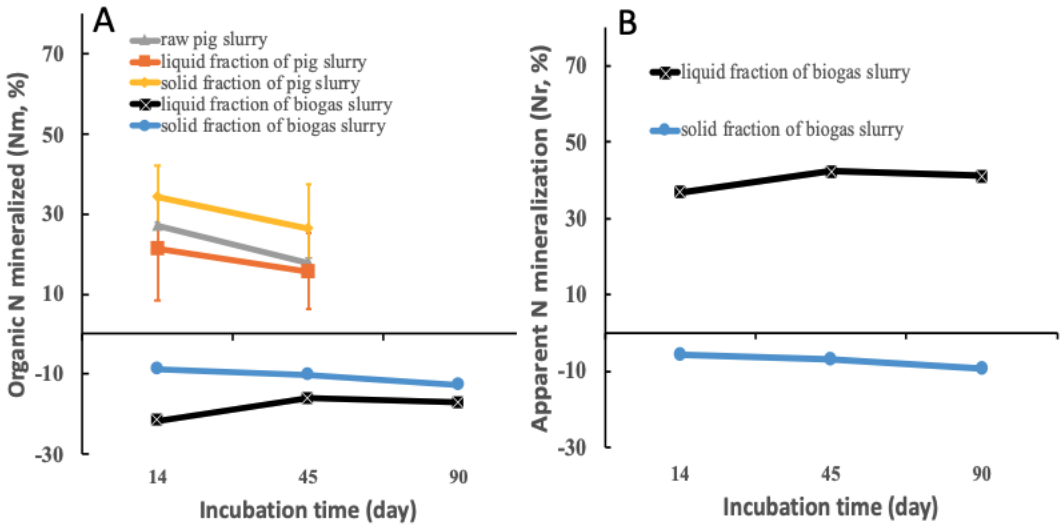



**Figure 4.8 Evolution of Nm(A) and Nr (B) during 90 days of incubation among slurry. Error bars indicate standard deviation.**

The pig slurry only observed Nm values on day 14 and 45 due to lack of data, but the trend of Nm followed a decrease pattern. Both solid and liquid fraction of biogas slurry showed net N immobilization during whole incubation period. On the other hand, the Nr of liquid fraction of biogas slurry remained positive around 40% across 90 days of incubation, mainly because 60% of total N was in mineral form.

### 4.1.3 Effects of BBF types, soil textures and time on N mineralization

Three-way ANOVA was carried out on a dataset with 23 individual experimental results to examine the effect of incubation day, BBF type and soil texture on Nm and Nr, respectively. There were in total 7 types of BBF and 8 types of soil texture included in the final dataset. However, due to the selection of data from publications with different experimental designs, some combinations of BBF types and soil textures were missing. For instance, digestates were only observed to apply on loamy sand soil and sandy loam soil in the dataset, but animal manures were applied to all soil textures included. The imbalanced datasets may cause difficulties to identify the interaction effects precisely between BBF types and soil textures. The calculated Nm represented the ability of organic N in BBFs mineralized into mineral forms. The value of Nr reflected the net mineral N release of BBFs at certain time. Statistical results were shown in Table 4.1.

**Table 4.1 Analysis of Variance for the effect of soil texture, BBF type and incubation day on Nm and Nr in a 3-way ANOVA model.**

| | Pr(>F) | |
|---|---|---|
| Independent variable | Nm ($R^2$=0.600) | Nr($R^2$=0.622) |
| Soil texture | <0.001 | <0.001 |
| BBF type | <0.001 | <0.001 |
| Incubation day | <0.001 | <0.001 |
| Soil texture*incubation day | 0.3079 | 0.1979 |
| BBF type* incubation day | <0.001 | 0.0476 |
| BBF type*soil texture | <0.001 | <0.001 |
| Incubation day* Soil texture*BBF type | 0.9905 | 0.9596 |

Note: the adjusted $R^2$ of each model was calculated from the reduced model with only statistically significant independent variables included.

The statistical results suggested that incubation days, BBF types and soil texture had significant effect on Nm. There was statistically significant interaction between BBF types and sampling time at 0.01 level, indicating that the relationship between BBF types and Nm was influenced by different sampling time (Table 4.1). Besides, the statistically significant interaction between BBF types and soil texture (Pr<0.001) implied that the effect of BBFs on Nm was influenced by soil texture as well. The 3-way interaction was not observed on Nm. As for the effects on apparent N mineralization (Nr), the effects of BBF types, incubation days, soil textures and the interaction between BBF type and soil texture were statistically significant. However, the interaction effect between incubation day and BBF types was no longer significant on Nr.

Figure 4.9 visualized the variation of Nm among BBF types at different time points. In general, the Nm of most BBFs did not exceed 50% through 90-day incubation period. On day 14, the percentage of organic N mineralized (Nm) among groups were not significantly different from each other, except for the slurry group. It seems that the Nm of slurry was significantly higher than the animal manure, compost and digestate groups, but the difference compared to green manures, industrial residues and sludges were no longer significant. The mean Nm was highest in slurry group at 21.1%, followed by sludge group with 8.4% of organic N mineralized within 2 weeks. The calculated Nm from the groups of composts and digestates varied in a relatively smaller range than the rest groups (Figure 4.9). In addition, compost and digestates tended to have negative value of Nm in most cases, which indicates that net immobilization was most likely to occur for these two types of BBFs. On day 45, the difference between animal manure and compost group became statistically significant ($p<0.05$). Except for that, the pairwise comparison indicated that the rest BBF groups released similar mineral N from organic sources. After 90 days of incubation, the statistical results indicated that both composts and digestates had significantly lower Nm than animal manures and industrial residues. Industrial residues outperformed with on average 27% organic N mineralized within 3 months, followed by animal manure with 17%. On the other hand, about 5% organic N mineralized in composts and the Nm of digestates tended to close to zero. It can be seen that the Nm of slurry group dropped sharply below zero on day 90. This extreme change was because the lack of data on Nm for several slurries on day 90: only the Nm of biogas slurry was available at this time point. The Nm of green manures showed greatest variation within groups on day 90, with the high Nm achieved by fresh white clover at 63.4% from Masunga *et al.* (2016) and low Nm of straw and cover crop mixture from Fontaine *et al.* (2020) at -15.4%, even though there were only 10 observations in this group.

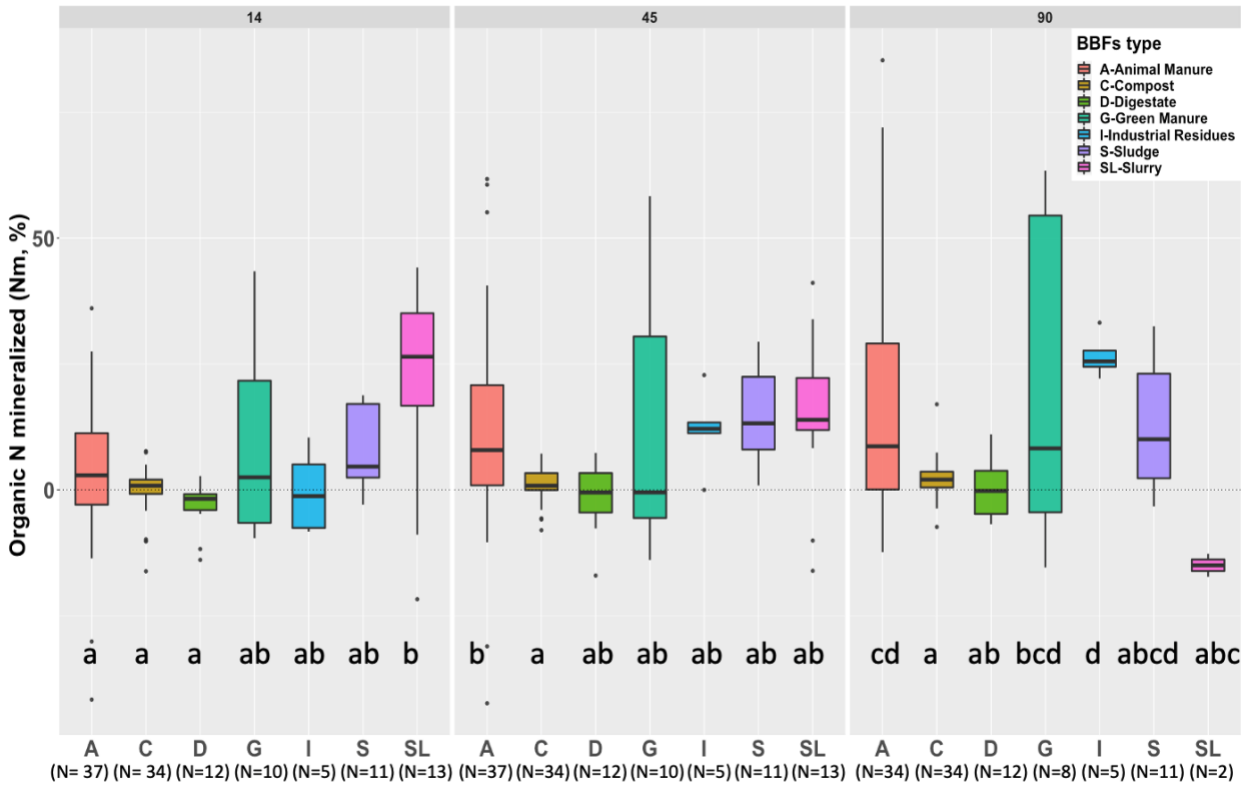



**Figure 4.9 Boxplot of Nm among BBF groups on different day. Lower case letters indicate the significant different estimated marginal mean of Nm by Tukey correction at 0.05 level per each incubation time (t=14, 45 and 90). A, animal manures; C, composts; D, digestates; G, green manures; I, industrial residues; S, sludge; SL, slurry.**

The apparent N mineralization (Nr) of BBFs on each incubation time (day 14, 45 and 90) is given in Figure 4.10. In general, the calculated Nr was fluctuated between 0 to 50% within 90 days, except for the digestate group. Just as the Nm of green manure, the Nr in green manure varied in a great range especially on day 90 from -11% of straw. (Fontaine *et al.*, 2020) to 64% of fresh white clover (Masunga *et al.*, 2016) with only 9 observations. By contrast, the Nr of compost group varied in a small range on day 90, from -5% of manure compost (Cardoso *et al.*, 2015) to 18% of household waste compost (Burgos, Madejón and Cabrera, 2006) with 36 observations. As mentioned earlier, the interaction between BBF type and incubation day had no significant effect on Nr, thus, the Nr differences among BBFs did not change by time. The estimated Nr mean of digestates (45%, 50% and 53% on day 14,45 and 90, respectively) was significantly higher than other BBF types, except for industrial residues. Besides, the slurry group was significantly different from digestates, but there were no clear differences detected between slurry and the rest groups. Meanwhile, composts resulted in significantly lower Nr compared to animal manures, digestates, industrial residues and sludges.

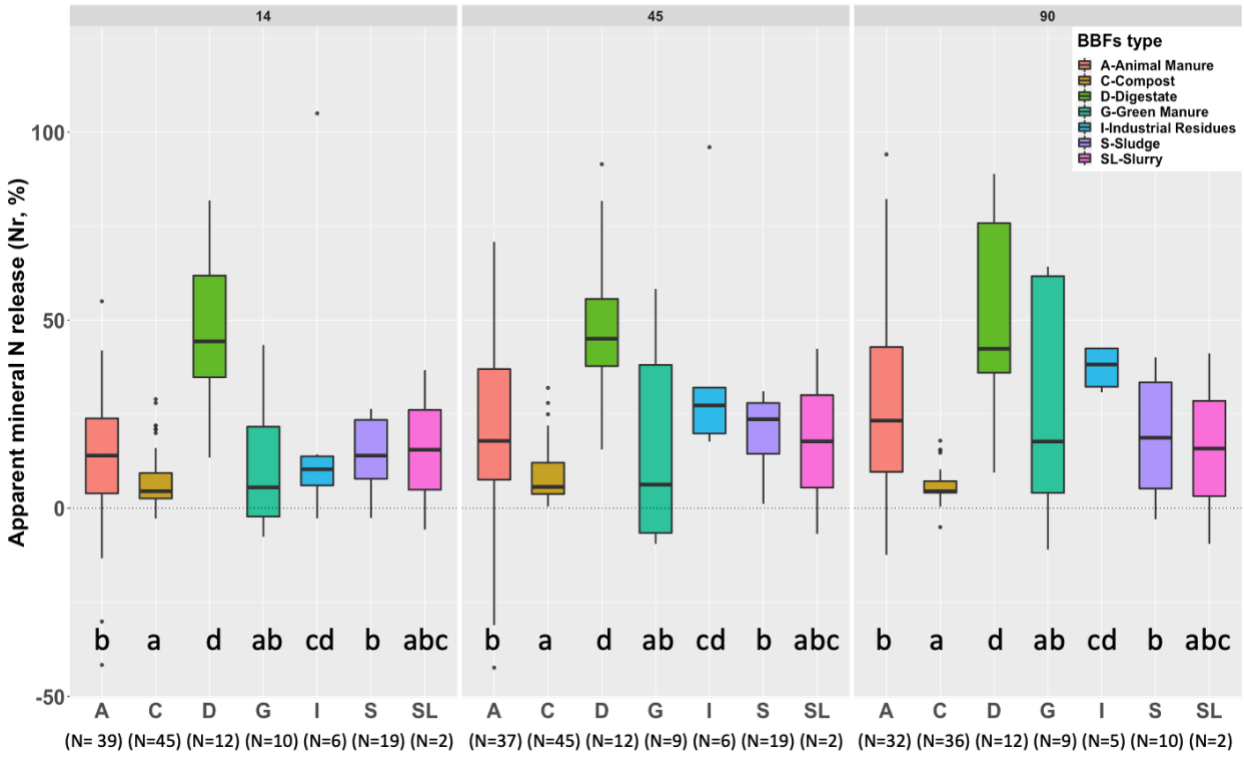



**Figure 4.10 Boxplot of Nr among BBF groups on different day. Lower case letters indicate the significant different estimated marginal mean of Nm by Tukey correction at 0.05 level per each incubation time (t=14, 45 and 90). A, animal manures; C, composts; D, digestates; G, green manures; I, industrial residues; S, sludge; SL, slurry.**

### 4.1.4 Effects of BBFs properties

In this section, the results of multiple linear regression (MLR) were presented on different sampling days (Table 4.2).

As can be seen in Table 4.2, the MLR model of Nm on day 14 was statistically significant ($p<0.001$), and the R-squared indicates that this set of predictors accounted for only 25.4% of the variations. A positive correlation was found with total C and organic N on Nm, whereas $NH_4$-N/total N ratio and C/N ratio were negatively correlated with Nm. As for Nr on day 14, the model explained 58.9% of the variability. $NH_4$-N/total N ratio had the strongest positive correlation to Nr with regression coefficient (RC) equals to 59.795. Total C also had significant effect on Nr, but with a considerably small RC (0.0422). On the other hand, C/N ratio (RC= -0.3756) showed a slight but significant negative effect on Nr.

On day 45, the independent variables included in MLR model varied slightly compared to day 14: the total N content of BBFs replaced organic N in the final model and was positively correlated with Nm. The impact of $NH_4$-N/total N ratio (RC=-40.8) and C/N ratio (RC=-0.45) remained dominant, whereas total N and total C content only explained small part of the variation on Nm. The influencing factor of Nr at day 45 was the same at day 14, with slightly change on the RC value of the independent variables: it seems that the negative effect of C/N ratio increased and at the same time positive effect of $NH_4$-N/total N ratio decreased.

On day 90, the BBFs properties that significantly influenced on Nm remained the same as the Nm on day 45. Nr at day 90 seemed to be mainly influenced by total C, C/N ratio and $NH_4$-N content at 0.001 significance level, but the effect of $NH_4$-N/total N ratio was no longer significant.

In general, Nr was well predicted by the model which explained around half of the variations under each incubation day. However, the MLR model for Nm did not fit well with relatively low R square value.

**Table 4.2 The significant explanatory variables (BBFs properties) on Nm and Nr on day14, 45 and 90 in MLR model.**

| Incubation day | 14 | | | 45 | | | 90 | | |
|---|---|---|---|---|---|---|---|---|---|
| Response Variable | Variable | Regression Coefficient | Significance Level | Variable | Regression Coefficient | Significance Level | Variable | Regression Coefficient | Significance Level |
| Nm | Total C | 0.0497 | ** | Total N | 0.10029 | ** | Total N | 0.18908 | * |
| | Organic N | 0.1936 | ** | Total C | 0.0459 | ** | Total C | 0.06049 | *** |
| | C/N ratio | -0.6818 | * | C/N ratio | -0.4486 | ** | C/N ratio | -0.51306 | ** |
| | $NH_4$-N/total N ratio | -16.4537 | * | $NH_4$-N/total N ratio | -40.758 | *** | $NH_4$-N/total N ratio | -57.91656 | *** |
| | $R^2$=0.3091, P<0.001 | | | $R^2$=0.244, P<0.001 | | | $R^2$=0.284, P<0.001 | | |
| Nr | Total C | 0.0422 | *** | Total C | 0.0808 | *** | Total C | 0.0782 | *** |
| | C/N ratio | -0.3756 | ** | C/N ratio | -1.277 | *** | C/N ratio | -0.6804 | *** |
| | $NH_4$-N/total N ratio | 59.795 | *** | $NH_4$-N/total N ratio | 41.767 | *** | $NH_4$-N | 0.4348 | *** |
| | $R^2$=0.6604, P< 0.001 | | | $R^2$=0.513, P<0.001 | | | $R^2$=0.489, P<0.001 | | |

Note: All variables included in the table are statistically significant at an alpha level of 0.05*, 0.01** or 0.001***.

### 4.1.5 Effects of soil properties

There was no interaction effect between soil textures and incubation days (Table 4.1), therefore, only Nm on day 90 was selected as response variable in the 3-way ANOVA analysis of soil properties. As indicated by the statistical results, soil temperature, pH and moisture content had significant effects on the Nm on day 90 in different significance level (Table 4.3).

**Table 4.3 the effect of soil properties on Nm in Multi-way ANOVA model.**

| response variable | variable | Sum of square | significance level |
|---|---|---|---|
| Nm on day 90 | soil pH | 23.021 | * |
| | soil temperature | 53.615 | ** |
| | soil moisture | 0.046 | * |
| | $R^2$= 0.3214, P<0.001 | | |

Note: The significance is at an alpha level of 0.01** and 0.05*.

The result of pairwise comparison illustrated that Nm on day 90 was significantly low when soil temperature was lower than 15ºC. When the soil moisture stayed at level 1(40%-50%), the Nm was significantly lower than at level 2(50-60%). It was noticed that the sum of square of soil moisture was lower than the rest two categorical variable, which indicated that the variation among moisture group was relatively small, though it significantly influenced Nm at 0.05 significance level. Moreover, the estimated marginal means of moisture group 1 and 3 had wide range of confidence interval, due to their less numbers of observations. The alkaline soil with pH higher than 7.5 resulted in significantly lower Nm than the neutral pH value (6.5-7.5).

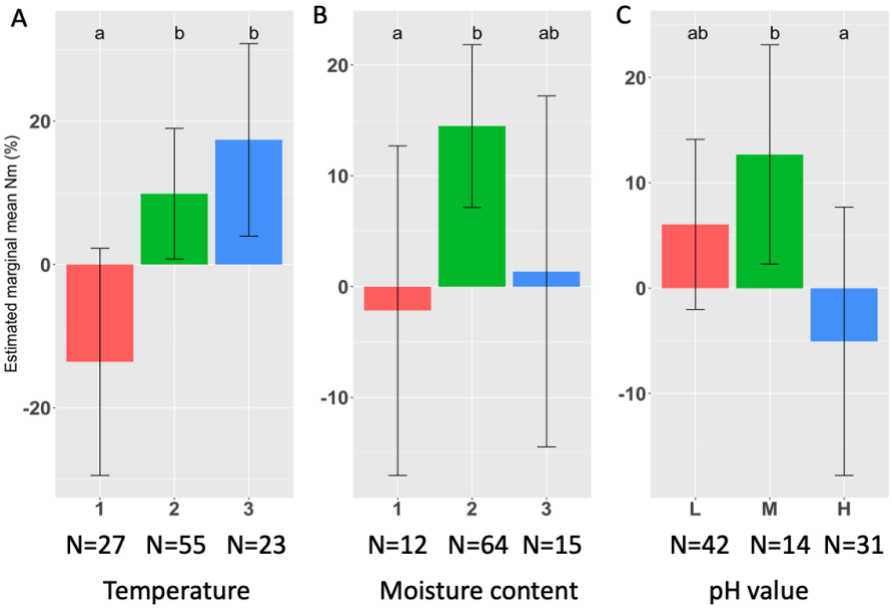



**Figure 4.11 The estimated marginal means of different groups for (A) Temperature, (B) Moisture content, and (C) pH value with 95% of confidence interval. Lower case letters indicate the significant different estimated marginal mean of Nm by Tukey correction at 0.05 level. Temperature groups: level 1 (T≤15°C), level 2 (15°C<T≤25°C), and level 3 (T>25°C); Moisture groups: level 1 (40%~50%), level 2 (50%-60%), and level 3 (>60%); pH groups: L (pH ≤6.5), M (6.5<pH≤7.5), and H (pH>7.5).**

## 4.2 Plant N uptake efficiency and N fertilizer replacement value of BBFs

### 4.2.1 General observations of NUE and NFRV from the dataset

The NUE values varied in a wide range from -11.8% to 82% in this dataset, being the lowest in straw treatment on spring barley (Fontaine *et al.*, 2020) and highest in pig slurry treatment on lettuce (Sigurnjak *et al.*, 2017a). The negative value on NUE indicated that the N uptake of spring barley from straw was lower than the control treatment without fertilizer application, which also implied that immobilization occurred to some extent during plant growth period. The results of ANOVA showed that both crop species and BBF types had significant effect on NUE at 0.01 significance level, with 80.34 % of variations explained by the model.

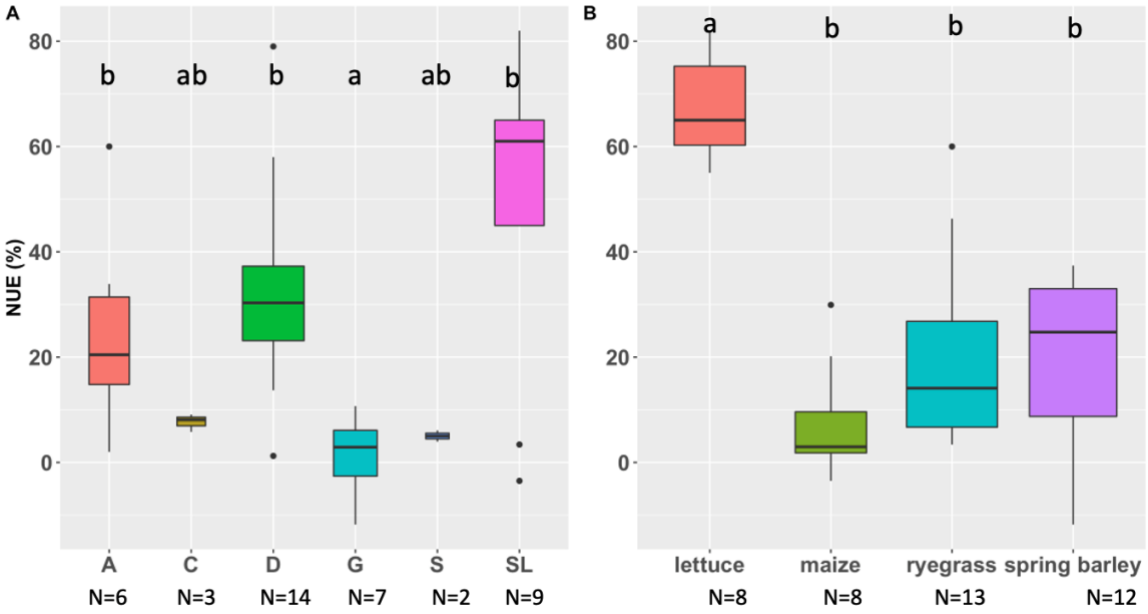



**Figure 4.12 Boxplot of NUE in (A) different BBF groups and (B) in different crops. Lower case letters indicate the significant different estimated marginal mean of Nm among groups by Tukey correction at 0.05 level.**

Among different BBF types, green manure showed significantly lower NUE than animal manure, slurry and digestates($p<0.05$). In the dataset, the lettuce with either digestate or slurry application by Sigurnjak *et al.* (2017a) tended to achieve higher NUE value than the rest crop with average 67% of total N in BBFs taken up by lettuce after harvested on day 54. By contrast, the NUE of maize with BBF application seemed to have relatively lower mean value at 5% for sewage sludge (Bozkurt *et al.*, 2006) and 25% for digestates (Cavalli *et al.*, 2016). There was a great variation on NUE when same type of BBFs applied on different crops from different literatures. For instance, NUE of maize from poultry manure in a pot experiment was only 2 % (Abbasi and Khizar, 2012).

By contrast, when poultry manure was applied for ryegrass in a pot experiment by Shah *et al.* (2013), the NUE value was significantly high at 60%.

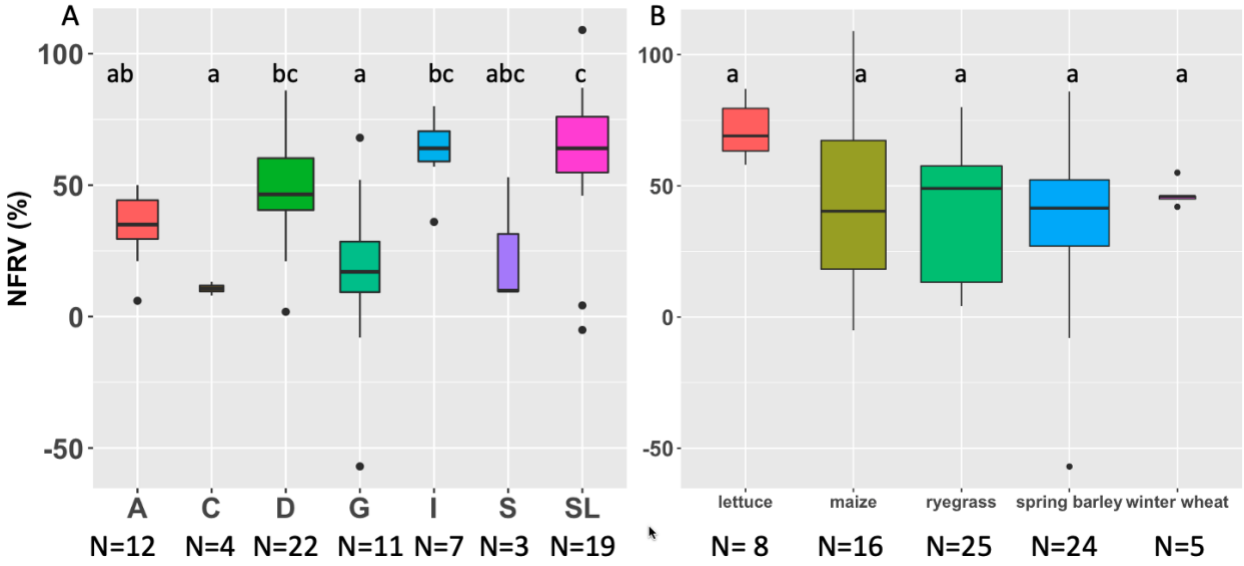



**Figure 4.13 Boxplot of NFRV in (A) different BBF groups and (B) in different crops. Lower case letters indicate the significant different estimated marginal mean of Nm among groups by Tukey correction at 0.05 level.**

The results of ANOVA on NFRV indicated that there was significant difference of NFRV among BBF types ($p<0.001$), but the difference among crop species was non-significant. Similar to the NUE results, green manures had significant lower NFRV than digestates and slurry group ($p<0.05$). There was a great variation of NFRV in the dataset. The highest NFRV value was observed for mineral concentrate of slurry at 109% in a field experiment on Maize (Schröder *et al.*, 2014), of which the fertilizer performance was even better than the synthetic fertilizer (calcium ammonium nitrate as the reference fertilizer in their study). However, the NFRV of straw treatment on spring barley from Fontaine *et al.* (2020) resulted in negative value at -57%, which was the lowest among the datasets. Cavalli *et al.* (2016) also recorded negative NFRV at -5.1% when the raw cattle slurry applied to maize in a field experiment, where the net N uptake of maize from the raw cattle slurry was 7 kg N/ha lower than control non-fertilizer group.

### 4.2.2 Effects of BBF properties on NUE and NFRV

The BBFs included in this dataset differed greatly on the chemical composition. The total N content varied between 4.7 g /kg DM in straw (Fontaine *et al.*, 2020), and 230 g /kg DM in the liquid fraction of pig slurry (Sigurnjak *et al.*, 2017a). Most of the BBFs in this dataset have N content lower than 100g/kg DM. Interestingly, the barley straw (Fontaine *et al.*, 2020) also had the highest C/N ratio at 99.78 and meanwhile the liquid fraction of pig slurry (Sigurnjak *et al.*, 2017a) had the lowest C/N ratio at 1.12. As for the $NH_4$-N/total N ratio, the lowest value was observed in blood meal from Delin *et al.* (2012) which contained only 0.6% $NH_4$-N of total N. However, the mineral concentrate of liquid pig slurry fraction held 100% of $NH_4$-N (Schröder *et al.*, 2013).

The results of linear regression indicated that total N, $NH_4$-N, C/N ratio and $NH_4$-N/total N ratio had significant effects on NUE and NFRV. As can be seen in Figure 4.14, the NUE values were positively correlated with total N content ($R^2$=0.5687; p<0.001) and negatively correlated with BBFs C/N ratio ($R^2$=05984; p<0.001), which both followed the logarithmic patterns. $NH_4$-N content per DM and $NH_4$-N/total N ratio were positively correlated with NUE, which were both fitted well with linear regression patterns. The regression model of $NH_4$-N/total N ratio on NUE achieved the highest $R^2$ value, where almost 70% of the variation could be explained by the model.

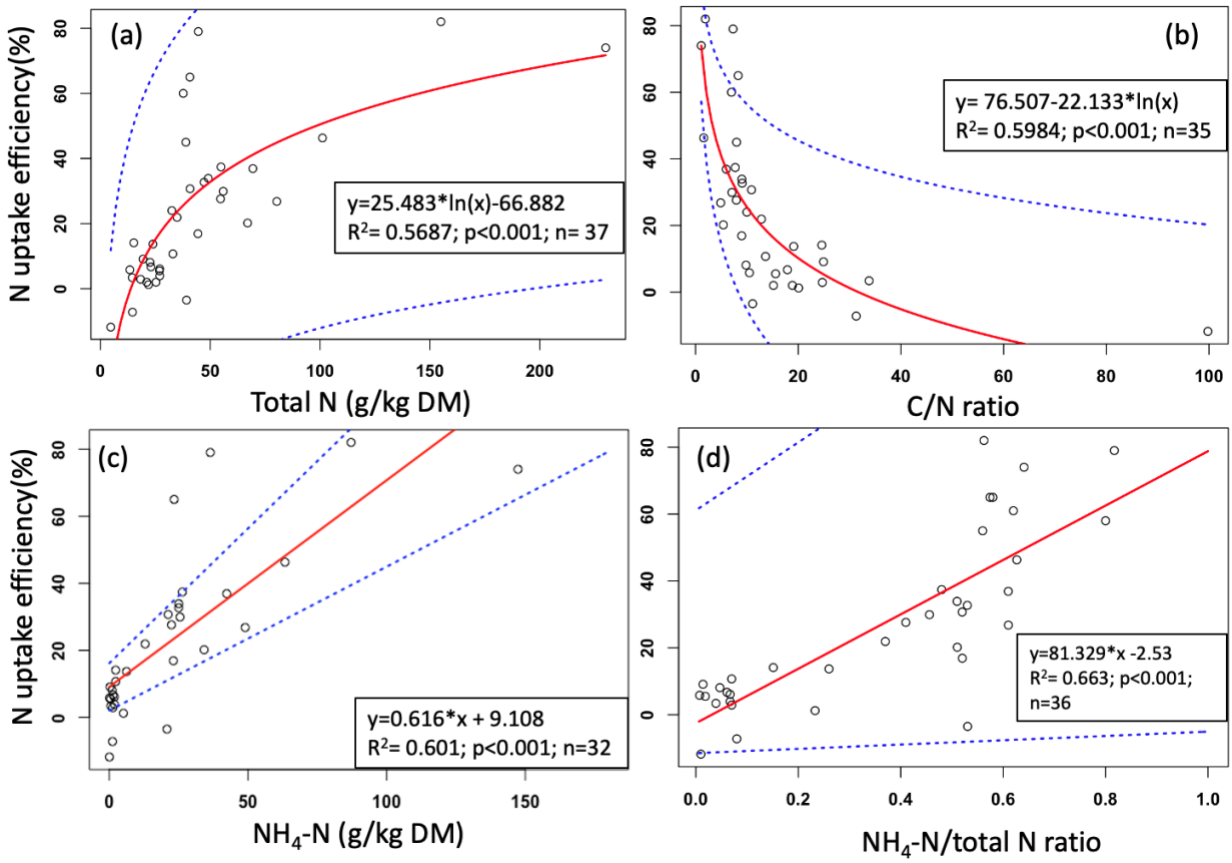



**Figure 4.14 The effect of (a) total N, (b) C/N ratio, (c) $NH_4$-N and (d) $NH_4$-N/total N ratio of BBFs on NUE. Red solid lines are predicted regression line based on the calculated correlation coefficient and blue dash lines indicated 95% of confidence interval.**

Similarly, the NFRV values were positively correlated with total N and $NH_4$-N content, and negatively correlated with C/N ratio of BBFs all in logarithmic transformations at 0.001 significance level (Figure 4.15). The relationship between $NH_4$-N/total N ratio and NFRV showed a positive linear pattern with relatively low $R^2$. The best correlation was observed for the negative effect of C/N ratio on NFRV, in which the linear model with log-transformed C/N ratio explained 72% of the variation in the dataset.

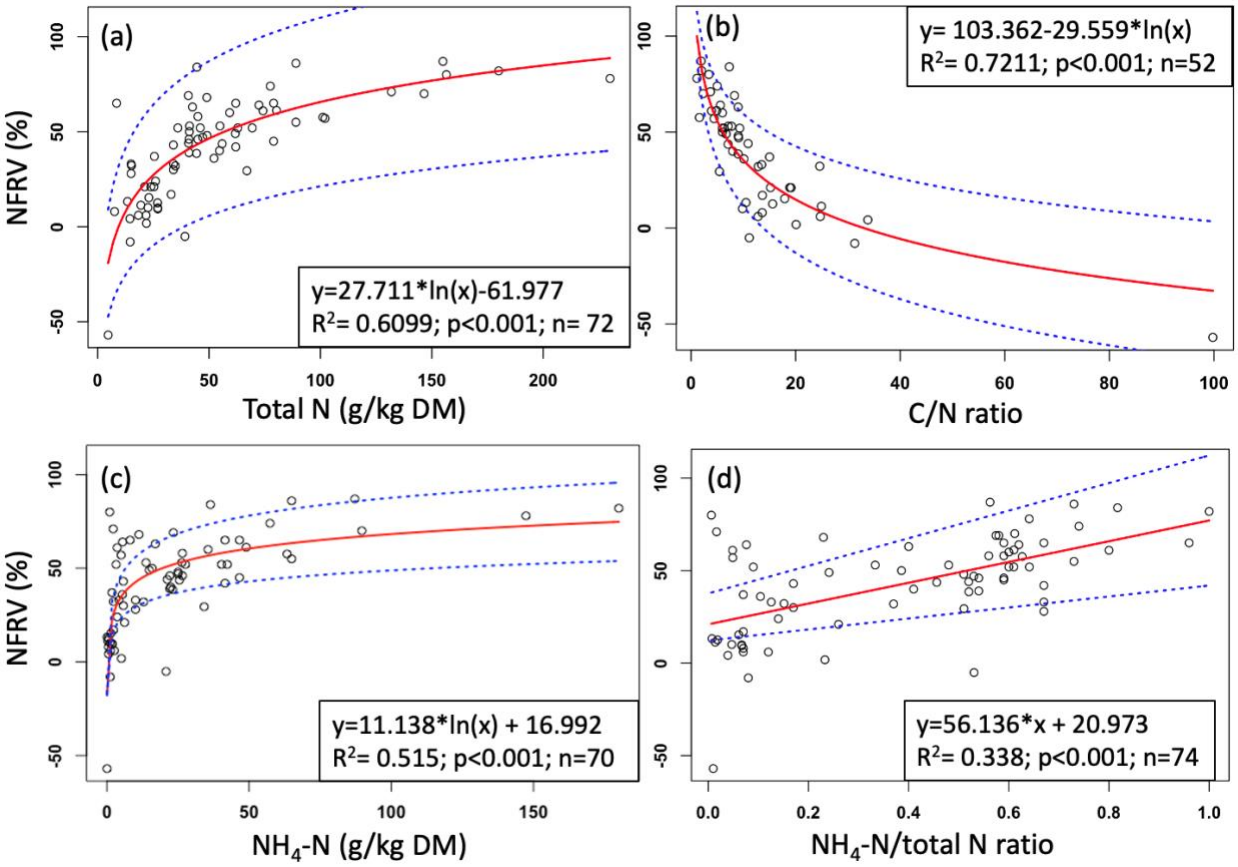



**Figure 4.15 The effect of (a) total N, (b) C/N ratio, (c) $NH_4$-N and (d) $NH_4$-N/total N ratio of BBFs on NFRV. Red solid lines are predicted regression line based on the calculated correlation coefficient and blue dash lines indicated 95% of confidence interval.**

# 5. Discussion

## 5.1 Influencing factors on BBFs N bioavailability

BBFs N bioavailability in this study is defined as the ability of BBFs offering plant available N (mineral N in $NH_4^+$ or $NO_3^-$ form) during crop growth period. The apparent mineral N release (Nr) and organic N mineralized (Nm) of BBFs were calculated as indicators to estimate N availability from different perspectives.

### 5.1.1 The N bioavailability among BBF types

The significant effect of BBF types on Nm and Nr in the 3-way ANOVA indicated that the composition of BBFs significantly influenced N bioavailability (Table 4.1). For instance, composts in this database had extremely lower $NH_4$-N/total N ratio (less than 0.1) than the digestates (average at 0.57). The most composts contained less than half of the total N compared to the digestates, which confirmed that treatment methods significantly influenced the total N content and $NH_4$-N proportion in BBFs (Pu *et al.*, 2012; Cavalli *et al.*, 2016). Thus, the digestates resulted in significantly higher Nr value than composts throughout the whole incubation period, which was mainly because of the high $NH_4$-N concentration in digestates (Figure 4.10). However, the Nm of composts and digestates were both lower than 10% and not significantly different from each other, which indicated that the organic N of both BBF types was recalcitrant for microbial degradation in a short-term. Case and Jensen (2019) also found that BBFs with high inorganic N /total N ratio tended to have negative Nm but positive Nr, mainly because their high $NH_4$-N content supplemented the mineral N in soil after fertilizer application. Therefore, it is reasonable to consider that digestates have high N bioavailability because of their high initial $NH_4$-N rather than the mineralization potential of digestates since most of the organic N in digestates is difficult to decompose in short period. Meanwhile, the low Nm rate of composts in this study could be also the result of highly stable materials retained in the composts that limited the ability of soil microbes mineralizing organic N within only 90 days of incubation. In previous studies, the organic N in composts were found to be more difficult to decompose in short term than the raw residues(Adegbidi and Briggs, 2003; Flavel and Murphy, 2006; Bechini and Marino, 2009). However, this is not always the case: Grigatti *et al.* (2011) found that the compost of solid biogas slurry fractions tended to have higher Nm than the raw solid fractions during the entire incubation period. The contrasting Nm pattern of composts was probably due to the difference on raw materials and composting process such as the choice of mixed materials, composting temperature and duration, which may alter the N concentrations and the degradability of organic matters

accordingly. Therefore, the bioavailability of BBF N was influenced by the choice of biological treatments and their corresponding technological processes.

The great variation of total N content in the slurry group indicated that solid and liquid separation process changes N composition of BBFs substantially: the liquid fractions of pig slurry from different separation technologies contained higher percent of total N (on average 27% in DM) and higher $NH_4$-N concentrations up to 92% of total N, compared to the corresponding solid fractions with only about 4% total N in DM and 24% of total N are in $NH_4$-N forms (Fangueiro *et al.*, 2014). The biogas slurry from Grigatti *et al.* (2011) showed similar trends on the change of N proportion after physical separation: the total N content of liquid fraction was about 10 times higher than the solid fraction of biogas slurry and the $NH_4$-N/total N ratio was significantly higher than in solid fraction. Therefore, upgrading treatments on raw organic residues are able to increase the short-term N availability of BBFs via the increasing share of total N and mineral N concentration in the final products, which was also proved by previous studies (Sradnick and Feller, 2020).

The pairwise comparison showed that the Nr value of animal manures and green manures were not significantly different from each other, even though the $NH_4$-N content of animal manures (mean at 14% of total N) was considerably higher than the green manures (mean at 5% of total N). Therefore, it is likely to be insufficient to evaluate N bioavailability only based on BBF's initial mineral N concentration, the organic N mineralization potential of BBFs should be estimated as well. Within the same BBF group, some certain BBFs showed distinct differences compared to others: the cattle manures in this study resulted in lower Nm value than poultry manures within 90 days of incubation (Figure 4.2), indicating a lower degradability of organic N in cattle manures. This was probably caused by the high concentration of fibre from forage and straw in cattle diet and resulted in cattle manures with a large proportion of resistant organic compounds such as cellulose and lignin (Sørensen and Fernández, 2003; Li and Li, 2014; Marzi *et al.*, 2020). The negative Nm value of barley straw from Fontaine *et al.* (2020) during 90 days of incubation also indicated that the agricultural-based lignocellulosic residues are difficult to decompose by soil microbes in short term. However, a study from Abbasi and Khaliq (2016) on wheat straw residue observed the re-mineralization of N on day 90, the process continued and reached highest mineral N release on the end day of incubation (day 140). The contrasting results could be explained by the differences on N concentration and the decomposability of organic residues in different crop species.

### 5.1.2 Effects of soil conditions on N availability

According to the statistical results, the effects of soil textures and the interaction between soil texture and BBF types on Nm and Nr were found to be significant (Table 4.1). However, as a consequence of data collection from different publications, the final dataset showed an imbalanced design and some combinations of BBF types and soil textures were missing. The imbalanced dataset causes difficulties on identifying interaction effects between soil textures and BBF types. More specifically, due to the lack of data, the statistical analysis fails to demonstrate how the effect of soil textures on Nm (or Nr) are influenced by different BBF types in this study. Moreover, the pairwise comparison among soil textures on Nm (or Nr) without considering the effects of BBFs seems unreasonable since the significant differences between two soil textures could be possibly only because different types of BBFs were applied. However, from an experimental study by Shah *et al.* (2013), the interaction effect between manure type and soil textures on Nr was significant: the Nr from both cattle manure and cattle slurry in their study were lower in the clay than in sandy soil ($P<0.05$), but for poultry manure no differences between soil textures were detected. In another incubation experiment on broiler litter, lower net N release was found when fertilizers were applied to higher clay content soil (Sistani *et al.*, 2008). These differences could be explained by the clay fixation of $NH_4^+$, which may stimulate microbial immobilization (Bechini and Marino, 2009). The high clay content in soil is also considered as physical prevention of organic N being mineralized by soil microbes (Stenger et al. 1995) and it may even stimulate denitrification because of high percentage of water-filled pore space in clay soils (Linn and Doran, 1984). The effect of soil textures on N mineralization were inconsistent across studies: Antil *et al.* (2011) argued that the effect of soil texture on net N mineralization was relatively small. Hadas and Portnoy (1994) also reported that soil texture had no significant effects on the decomposition rate of different manure composts.

The short-term N mineralization is biologically determined by soil microbial activities and different soil conditions. The effects of soil temperature, moisture content and pH on microbial activities are frequently discussed by previous studies(Kemmitt *et al.*, 2006; Huang and Chen, 2009; Zare and Ronaghi, 2019). The statistical results in this study showed that low temperature tended to result in low N mineralization rate (Figure 4.11), which could be the reason that lower temperature may inhibit microbial activity and cause low decomposition rate on organic N in soil. Previous studies found that gross N mineralization tended to increase with the increasing soil temperature, meanwhile the higher temperature also stimulated gross N immobilization due to the growth of microbial community (Andersen and Jensen, 2001; Thangarajan *et al.*, 2015; Miller and Geisseler, 2018). Moreover, Cassity-Duffey *et al.* (2018) found the significant interaction effect between manure types and temperature on net N mineralization, which was also argued by

Whitmore (2007), who stated that the hardly degradable components are more affected by temperature than the easily degradable organic matters in BBFs. Soil moisture content was also found to significantly influence Nm in this study. However, the sample size for moisture level 1(40-50%) and 3(>60%) were too small compared to level 2(50-60%), resulting in a wide range of confidence interval (Figure 4.11), which may cause difficulties to provide valid information on moisture sensitivity of net N mineralization. The significantly low Nm at high pH level group (>7.5) might be because potential increase of ammonia volatilization or the unfavorable condition of microbial activities(Amlinger *et al.*, 2003). Overall, the effects of soil temperature, moisture and pH on organic N mineralization were found to be statistically significant, but the interpretation on their effects from a soil microbiological perspective remained unclear in this study.

## 5.2 Effects of different BBF properties on N mineralization

Multi-linear regression (MLR) analysis was conducted to identify the effects of BBF properties on Nm and Nr, respectively. The statistical results showed that $NH_4$-N/total N ratio had dominant negative effects with the highest absolute value of regression coefficient (RC) on Nm in regardless of sampling time (day 14, 45 and 90). On the other hand, the results on Nr indicated that $NH_4$-N/total N ratio had strongly positive effect in the early and middle stage of incubation (day 14 and 45). The opposite effects of $NH_4$-N/total N ratio on Nm and Nr must have been because the different parameters used in the calculation (Table 3.6): Nm represented only the proportion of organic N mineralized from BBFs to total N application rate. In contrast, Nr denoted the sum of organic N mineralized and the initial mineral N in BBFs as the percentage of total N applied. Therefore, BBFs with higher $NH_4$-N/ total N ratio implies that fewer organic N in BBFs could be decomposed by soil microbes in the first place, which may cause the Nm decrease with the increase of BBF $NH_4$-N concentration. Fontaine *et al.* (2020) observed the significant positive effect of $NH_4$-N/total N ratio on Nr for digestates in a 180-day incubation study, though it was not significant for undigested residues. It is noticed that on day 90, the correlation between $NH_4$-N/total N ratio and Nr became insignificant, and it was replaced by $NH_4$-N content (g/kg DM) to be positively correlated with Nr (Table 4.2). It could be argued that the cumulative mineral N in BBF applied soil (Nr) after a relatively long term were influenced more by the amount of $NH_4$-N in BBFs per DM rather than the $NH_4$-N/total N ratio. Similar results were found in Sigurnjak *et al.*(2017b) , where $NH_4$-N/total N ratio of BBFs had no significant effect on Nr but $NH_4$-N content was detected to be positively correlated with Nr after 120 days of incubation.

C/N ratio was found to be negatively correlated with Nm and Nr at each time point in all MLR models (Table 4.2). Previous studies have well documented on the significant effect of C/N ratio on N mineralization (Qian and Schoenau, 2002; Flavel and Murphy, 2006; Morvan, Nicolardot and Péan, 2006; Dossa *et al.*, 2009; Rigby *et al.*, 2016). Joseph *et al.* (2017) analyzed the effect of C/N ratio on Nr based on the data collected from literatures and pointed out that Nr decreased with the increasing C/N ratio of BBFs. They also found that when C/N ratio is higher than 38, net N immobilization is likely to occur. In fact, the literatures (Grigatti *et al.*, 2011; Antoniadis, 2013; Fontaine *et al.*, 2020) included in this database also observed the negative effect of C/N ratio on Nm or Nr. However, contradict results were observed in Li and Li (2014) and Nurdiawati *et al.* (2019), where C/N ratio had no significant correlation with N mineralization in soil after fertilizer application. This may have been because the inclusion of BBFs in their study had similar C/N ratio which makes other chemical properties such as organic N or physical properties more determine on N mineralization (Burger and Venterea, 2008; Fangueiro *et al.*, 2014; Nurdiawati *et al.*, 2019).

Although the C/organic N ratio was found to be non-significant in all MLR models in this work, other studies did observe that C/organic N ratio of BBFs significantly influenced N mineralization/immobilization process (Peters and Jensen, 2011; Sradnick and Feller, 2020).

Organic N of BBFs was found to be positively correlated with Nm on day 14 but the effect of total N was not significant, which implied that organic N mineralization process was likely to be affected by organic N rather than total N content of BBFs in the beginning of incubation. However, total N content was found to be significantly related to Nm in the middle and later stage of incubation (day 45 and 90), but organic N of BBFs showed insignificant effect and thus was excluded in the MLR model (Table 4.2). Mubarak *et al.* (2010) found that total N content of manure was positively correlated to Nm after 84 days of incubation experiment. Galvez *et al.* (2012) also found that Nm at the end of the incubation (day 70) was significantly correlated to the total N content of BBFs. By contrast, neither total N nor organic N were included in the MLR model on Nr, which could be because the effect of $NH_4$-N/total N ratio on Nr was more dominant and caused the insignificance of total N in the MLR model.

Total C content of BBFs showed significant effect on Nm and Nr, though the RC values were less than 0.1 at all sampling times. However, no selected literature recorded the significant effect of total C on Nm or Nr. Other studies observed the effect of C on N mineralization: Flavel and Murphy (2006) conducted $^{15}N$ isotopic pool dilution experiment and discovered that total C content of BBFs was positively correlated with gross N mineralization; Partey, Preziosi and Robson (2014) found that the levels of extractable C of BBFs were significantly correlated to N availability in soil.

## 5.3 Plant N uptake efficiency and N fertilizer replacement value of BBFs

### 5.3.1 Effects of BBF properties

The statistical results on simple linear regression showed that total N, $NH_4$-N and $NH_4$-N/total N ratio had significantly positive effects on first year NUE (Figure 4.14), which was consistent with the findings from several selected literatures (Grigatti *et al.*, 2011 and Benke *et al.*, 2017) in this dataset. A very recent study from Pedersen *et al.* (2021) also observed that $NH_4$-N/total N ratio of pig slurry positively correlated with first year plant NUE. Therefore, BBFs with both higher total N content and $NH_4$-N/total N ratio tend to achieve higher plant NUE in the first year of fertilizer application. This was also observed among BBF types that included in this study (Figure 4.12, A): slurry and digestate group tended to achieve higher NUE than animal manures or composts, mainly because of their higher $NH_4$-N/total N ratio. The C/N ratio, on the other hand, was found to be negatively correlated with NUE, which was also observed in other studies (Frøseth *et al.*, 2014; Benke *et al.*, 2017; Joseph *et al.*, 2017). It is noticed that the performance of NUE should not be solely determined on only one BBF property. For instance, in Figure 4.14-(c), there were 4 data points (the BBFs) from Sigurnjak *et al.*(2017a) that achieved relatively high NUE (65-82%) but had considerably different $NH_4$-N content. When traced back to the dataset, these 4 BBFs actually had similar and high $NH_4$-N/total N ratio at around 0.6, which explained their high performance on short-term NUE value.

The first year NFRV was positively correlated to total N and $NH_4$-N/total N ratio (both $p<0.001$) in this study (Figure 4.15), and the same patterns were observed for NUE as well. The literatures included in this dataset also observed the significantly positive effects of total N (De Notaris *et al.*, 2018) and $NH_4$-N/total N ratio (Schröder *et al.*, 2013; De Notaris *et al.*, 2018; Fontaine *et al.*, 2020) on NFRV in their own experiments. The $NH_4$-N content fitted well with a logarithmic pattern to NFRV (Figure 4.15) and showed a different increasing pattern compared to the $NH_4$-N to NUE model (Figure 4.14). This could be because of the imbalanced dataset that the NFRV had more observations than NUE and most data points scattered within a small range (0 and 40 g/kg DM). Previous literatures argued that C/N ratio had significantly negative effect on first year NFRV (Sørensen and Fernández, 2003; Delin *et al.*, 2012; Fontaine *et al.*, 2020), which was also confirmed by the statistical results in this study. Moreover, the regression model of C/N ratio on NFRV had the best performance and explained 72% of the variation in the statistically analysis, which could be argued that C/N ratio is a good estimator on the first NFRV in a general term. On the other hand, the regression model of $NH_4$-N/total N ratio on NFRV explained less than half of

the variability, even though the effect was statistically significant. Fontaine *et al.* (2020) observed that the significant effect of $NH_4$-N/total N ratio on NFRV was relatively lower for undigested materials compared to digestates. This could have been because the effect of different fertilizer types: the influence of $NH_4$-N/total N ratio on NFRV should be more dominant for digestates and liquid slurry, since they normally contain high $NH_4$-N concentration. On contrary, other parameters such as C/N ratio, total N content and amount of easily degradable materials seem to be more determining on short-term NFRV, when the BBFs are at low $NH_4$-N/total N ratio.

### 5.3.2 Other possible influencing factors on plant NUE and NFRV

Apart from the BBFs chemical properties, several external factors that may influence NUE and NFRV were discussed in previous studies. In the field experiments, the weather conditions partly contribute to the variability of NFRV (Schröder *et al.*, 2013). Besides, different application method and time could result in different NFRV value (De Notaris *et al.*, 2018). Those influencing factors are mainly attributed to soil microbial activities, the extent of $NH_3$ volatilization and $NO_3^-$ leaching on fields after BBFs application. For instance, Delin (2011) found that NUE of spring barley with broiler manure by shallow incorporation was higher in early spring, which may be because the lower temperature resulted in lower ammonia emission than the application at sowing. Similarly, De Notaris *et al.* (2018) observed that when BBFs were applied before sowing via soil injection, the anomia emission was reduced and meanwhile the organic N mineralization process can synchronize better with the N demand of crop growth later. Lalor *et al.* (2011) compared the NFRV of slurry between trailing shoe and splash plate application. They found that trailing shoe achieved higher NFRV value which was mainly due to the lower ammonia emission during slurry application.

It is needed to mention that the calculation of NFRV in this study is based on comparison between the crop N uptake efficiency on BBF and the mineral N fertilizer. As mentioned in the literature review, the NFRV value can be also calculated based on DM yields (Table 2.1), which is easier to evaluate (since the N uptake of crops involved with chemical analysis on biomass) and seems more relevant to farmer's concerns (Jensen 2013; Schröder 2005). The calculated NFRV of the same BBF could be different when different references are used (De Notaris *et al.*, 2018) and it is important to keep that in mind when comparing the NFRV values to other studies.

## 5.4 Discussions on Methodology

The selected 23 literatures for the meta-analysis on N mineralization potential were all laboratory incubations, where BBFs were thoroughly mixed with soil and experimental conditions on soil temperature and moisture were constant. Strictly speaking, the results from lab experiments may not accurately reflect the mineralization process on field in different climate conditions. Compared to a similar research by Sradnick and Feller (2020) with as much as 835 observations on BBFs N mineralization, the dataset in this study was relatively small, with only 144 data points. Besides, the BBF types were roughly categorized based on the sources and treatment methods, which caused some groups having relatively high variations on the chemical properties (e.g., slurry group). More specified categorization could have been applied, for instance, classified further according to BBFs forms (liquid or solid). The response variables Nm and Nr were all calculated as the proportion of total N applied (%). This data normalization method aimed to make the results from different literatures comparable, while this process also ignored the effect of BBF application rates on N mineralization potential. Besides, the percentage of organic N mineralized (Nm) was calculated based on total N content in order to include more observations in the statistical analysis. It could be more suitable to calculate Nm based on organic N content because some BBFs such as digestates and slurry generally contain high amount of initial inorganic N. Similarly, the NFRV value was scaled to the percentage of certain mineral N fertilizer that can be replaced by BBFs. However, as can be seen in the dataset (Appendix 3), different mineral N fertilizers were applied, and plants may achieve different NUE in the first place. Therefore, a standardized method of calculating NFRV should be established, probably first to unify the reference mineral N fertilizer to make different experimental results more comparable.

# 6. Conclusions

The main objective of this study is to understand the ability of BBFs offering plant available N in a short-term period, which was elaborated by answering the three research questions:

(1) *Which factors influence the bioavailability of N in soil after BBF application?*

This study found that BBF types, soil textures and time significantly influence N bioavailability in soil after application. Different BBFs vary greatly in physical and chemical compositions, which are regulated by the sources of organic residues and treatment methods before application. Soil properties such as temperature, moisture and pH influence the N dynamic as well, mainly by altering the activity of soil microbes on mineralization process and the extent of ammonia volatilization. It is observed that mineral N release of BBFs change with time, which indicated that N bioavailability of BBFs could perform differently in short-term and long-term period. In this study, composts showed lower short-term N bioavailability than digestates, but their ability on gradually releasing mineral N in a long term should not be underestimated.

(2) *Which properties are related to N mineralization on releasing available N in different types of BBFs?*

The total C, total N, organic N, $NH_4$-N, $NH_4$-N/total N ratio and C/N ratio of BBFs showed statistically significant effects on N mineralization potential, in which the effect of $NH_4$-N/total N ratio and C/N ratio were found to be more dominant. From the reviewed literatures and the results of meta-analysis, it is suggested that BBFs tend to perform better on short-term N release in two conditions: 1) When BBF contains higher total N or organic N content but lower C/N ratio; or 2) When BBFs contains higher $NH_4$-N share of total N but lower C/N ratio. In sum, the short-term N availability of BBFs should not be evaluated only according to one property, various information on BBF property is required to estimate the optimal application rate of certain BBFs on field.

(3) *Is the plant N uptake efficiency (NUE) or nitrogen fertilizer replacement value (NFRV) of BBFs related to the same factors and properties that influence N mineralization potential?*

In this study, total N, $NH_4$-N, $NH_4$-N/total N ratio and C/N ratio of BBFs had statistically significant effects on NUE and NFRV. $NH_4$-N/total N ratio had the best fit with NUE in a positive linear pattern and the correlation between C/N ratio and NFRV was found to be the strongest. Considering the results on NUE and NFRV together with Nm and Nr, it could be concluded that the BBF with higher $NH_4$-N /total N ratio and lower C/N ratio is likely to achieve higher short-term N availability and thus results in higher NFRV value for the first-year application.

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# 8. Appendices

**Appendix 1 The minimum, maximum, mean, median and standard deviation of each chemical property on each BBF group.**

| BBF types | BBF properties | | | | |
|---|---|---|---|---|---|
| | total N content g/kg DM | | | | |
| | minimum | maximum | mean | median | standard deviation |
| Animal manure | 6 | 65.9 | 25.7 | 23.4 | 12.1 |
| Compost | 9.6 | 49 | 29.9 | 34.5 | 12.5 |
| Digestate | 23.9 | 209 | 82.2 | 58.2 | 61.8 |
| Green manure | 4.7 | 63 | 26.5 | 24 | 17.1 |
| Industrial residue | 17.4 | 79.8 | 36.2 | 30 | 22.1 |
| Sludge | 2.74 | 60.6 | 43.9 | 54.1 | 20.1 |
| Slurry | 14.5 | 319 | 141 | 63.1 | 120 |
| | total C content g/kg DM | | | | |
| | minimum | maximum | mean | median | standard deviation |
| Animal manure | 41.4 | 443 | 296 | 331 | 107 |
| Compost | 63 | 483 | 226 | 196 | 93.3 |
| Digestate | 263 | 457 | 388 | 420 | 66.5 |
| Green manure | 375 | 469 | 428 | 434 | 33.5 |
| Industrial residue | 137 | 417 | 327 | 343 | 106 |
| Sludge | 162 | 485 | 357 | 364 | 70.4 |
| Slurry | 161 | 490 | 277 | 260 | 79.9 |
| | C/N ratio | | | | |
| | minimum | maximum | mean | median | standard deviation |
| Animal manure | 3.6 | 35 | 13.4 | 12.1 | 7.29 |
| Compost | 5.68 | 24.9 | 8.39 | 7.1 | 4.33 |
| Digestate | 1.26 | 19.1 | 7.48 | 7.01 | 5.12 |
| Green manure | 7.33 | 99.8 | 27.7 | 16.4 | 27.9 |
| Industrial residue | 1.72 | 21.8 | 11.9 | 12 | 6.39 |
| Sludge | 6.01 | 94.9 | 14.3 | 6.3 | 20.3 |
| Slurry | 0.8 | 33.8 | 5.8 | 4.5 | 8.82 |
| | $NH_4$-N/total N ratio | | | | |
| | min | max | mean | median | standard deviation |
| Animal manure | 0.0127 | 58.9 | 13.9 | 7 | 16.7 |
| Compost | 0.122 | 13.9 | 8.5 | 13.9 | 6.53 |
| Digestate | 26 | 95 | 57 | 52.5 | 19.6 |
| Green manure | 1 | 8 | 5.75 | 7 | 3.2 |
| Sludge | 0.423 | 10 | 6.53 | 8.12 | 3.73 |
| Slurry | 3.92 | 92 | 44.8 | 30.9 | 29.7 |

**Appendix 2 Main R code for statistical analysis.**

<table>
<tr><th colspan="2">BBFs N mineralization</th></tr>
<tr><td>(1) 3-way ANOVA<br>m1<-lm(Nm~day*BBF*s_type,data=Nmnoend) anova(m1)<br>drop1(m1,test="Chisq")<br>drop1(m2<update(m1,.~.day:BBF:s_type),test="Chisq")<br>drop1(m3<-update(m2,.~.-day:s_type),test="Chisq")<br>drop1(m4<-update(m3,.~.-BBF:day),test="Chisq")<br>anova(m3)<br>summary(m3)<br>emmeans(m3, pairwise ~ BBF | day)<br><br>mr1<-lm(Nr~day*BBF*s_type,data=Nrnoend)<br>anova(mr1)<br>drop1(mr1,test="Chisq")<br>drop1(mr2<-update(mr1,.~. day:BBF:s_type),test="Chisq")<br>drop1(mr3<-update(mr2,.~.-day:s_type),test="Chisq")<br>drop1(mr4<-update(mr3,.~.-BBF:day),test="Chisq")<br>anova(mr4)<br>summary(mr4)<br>emmeans(mr4, pairwise ~ BBF | day)</td><td>(2) MLR<br>m14_nm<-lm(Nm_14~B_TN+B_TC+B_ON+B_CN+B_C_ON+B_NH4_TN+B_NH4 , data=sub14_nm)<br>m45_nm<-lm(Nm_45~B_TN+B_TC+B_ON+B_CN+B_C_ON+B_NH4_TN+B_NH4 , data=sub45_nm)<br>m90_nm<-lm(Nm_90~B_TN+B_TC+B_ON+B_CN+B_C_ON+B_NH4_TN+B_NH4 , data=sub90_nm)<br>m14_nr<-lm(Nr_14~B_TN+B_TC+B_ON+B_CN+B_C_ON+B_NH4_TN+B_NH4 , data=sub14_nr)<br>m45_nr<-lm(Nr_45~B_TN+B_TC+B_ON+B_CN+B_C_ON+B_NH4_TN+B_NH4 , data=sub45_nr)<br>m90_nr<-lm(Nr_90~B_TN+B_TC+B_ON+B_CN+B_C_ON+B_NH4_TN+B_NH4 , data=sub90_nr)<br><br>(3) Soil properties<br><br>M_s<-( Nm_90~s_temp+s_moisture+s_ph, data= sub90_nm)</td></tr>
<tr><th colspan="2">NUE and NFRV</th></tr>
<tr><td>(1) 2-way ANOVA<br>r1<- lm (NFRV~BBF+crop,data=df)<br>r2<- lm (NUE~crop+BBF,data=df1)</td><td>(2) Simple linear regression<br>x1<-lm(NFRV~log(TN),data= NFRV) fit<-lm(NUE~log(TN),data=NFRV)<br>x2<-lm(NFRV~log(CN),data= NFRV) fit1<-lm(NUE~log(CN),data=NFRV)<br>x3<-lm(NFRV~NH4TN,data= NFRV) fit2<-lm(NUE~NH4TN,data=NFRV)<br>x4<-lm(NFRV~log(NH4),data= NFRV) fit3<-lm(NUE~NH4,data=NFRV)</td></tr>
</table>

**Appendix 3 General information of the literatures selected for semi- quantitative analysis of N uptake efficiency and N fertilizer replacement value.**

| Reference | Location | Crop | Experiment | Application method | Mineral N fertilizer | Bio-based fertilizer treatment |
|---|---|---|---|---|---|---|
| Abbasi and Khizar. 2012 | Pakistan | Maize | pot experiment | by hand | urea N | poultry manure; white clover residue; poultry manure+ white clover residue mixture |
| Benke et al. 2017 | Germany | Ryegrass | pot experiment | by hand | calcium ammonium nitrate | fresh clover grass biomass; clover grass silage; composted clover grass; biomass; solid dairy manure; liquid dairy manure; digestates based on clover grass |
| Bozkurt et al., 2006 | Turkey | Maize | field experiment | by hand (incorporated) | ammonium sulphate | sewage sludge |
| Cavalli et al. 2016 | Italy | Maize | field experiment | trailing hose spreader; by hand | ammonium sulphate | unseparated digestates of cattle slurry and maize mixture (DSMM); liquid fraction of DSMM; solid fraction of DSMM; untreated cattle slurry |
| Delin et al. 2011 | Sweden | Spring barley | field experiment | shallow incorporation | - | broiler manure; hen manure; broiler manure; hen manure |
| Delin et al. 2012 | Sweden | Ryegrass | pot experiment | by hand | ammonium nitrate | Blood meal; Bone meal; Feather meal; Horse manure; Meat meal; Chicken manure; Lucerne pellets; Mink manure; Mussel compost; Cattle slurry; Rapeseed cake; Biogas residue; Sewage sludge; Pig slurry; Vinasse |
| De Notaris et al. 2018 | Denmark | Winter wheat; spring barley | field experiment | surface-banding or injection | ammonium nitrate | green manure; green manure digestate; raw cattle slurry; digestate cattle slurry; raw cattle slurry |

| | | | | | | |
|---|---|---|---|---|---|---|
| Fontaine et al. 2020 | Denmark | Spring barley | field experiment | - | ammonium nitrate | green manure, cattle manure, green manure and cattle manure mixture; digestates; straw |
| Grigatti et al. 2011 | Italy | Ryegrass | pot experiment | by hand | urea N | liquid fraction and solid fraction from biogas slurry; composted solid fraction of biogas slurry; municipal solid waste compost |
| Schröder et al. 2013 | Netherlands | Maize | field experiment | Liquid fertilizer injection; solid manure spreader | calcium ammonium nitrate | liquid fraction from separated pig slurry; untreated pig slurry ; untreated cattle slurry; solid fraction of separated pig slurry; farmyard manure |
| Schröder et al. 2014 | The Netherlands | Maize | field experiment | injected | calcium ammonium nitrate | mineral concentrate of slurry |
| Shah et al. 2013 | Netherlands | Perennial Ryegrass | pot experiment | by hand | none | poultry manure; cattle slurry; solid cattle manure |
| Sigurnjak et al. 2017a | Belgium | Lettuce | pot experiment | - | calcium ammonium nitrate | pig slurry; liquid fraction of pig slurry digestate; liquid fraction of digestate |

Note: the more detailed information on each literature can be found in supplement material 2.